

SCIENTIFIC EXPLORATION OF EXTRATERRESTRIAL PLANETS AND ANALYSIS USING QUANTUM COMPUTING

An Undergraduate CAPSTONE Project
By

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**Fall Semester 2019-2020,
December, 2020**



**Faculty of Engineering
American International University - Bangladesh**

SCIENTIFIC EXPLORATION OF EXTRATERRESTRIAL PLANETS AND ANALYSIS USING QUANTUM COMPUTING

A CAPSTONE Project submitted to the Faculty of Engineering, American International University - Bangladesh (AIUB) in partial fulfillment of the requirements for the degree of Bachelor of Science in their mentioned respective programs.

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**Faculty of Engineering
American International University - Bangladesh**

DECLARATION

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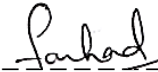
APPROVAL

The CAPSTONE Project titled **SCIENTIFIC EXPLORATION OF EXTRATERRESTRIAL PLANETS AND ANALYSIS USING QUANTUM COMPUTING** has been submitted to the following respected members of the Board of Examiners of the Faculty of Engineering in partial fulfillment of the requirements for the degree of Bachelor of Science in the respective programs mentioned below on **December, 2019** by the following students and has been accepted as satisfactory.

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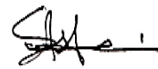
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ABSTRACT

This project presents a simplified model of a Mars Exploration Rover whose purpose is to serve as the 'eyes and brain' of a planetary scientist. This model focuses on machine vision and scientific instrumentation for effective in-situ analysis in advanced rover technology and experimental processing through quantum computation. For the vision aspect, images of the Martian terrain were primarily used to isolate, detect and classify geological features (i.e., rocks) through mask-edge filtering, SNR desolution, and rendered output, yielding an average of 88.8% accuracy in detection and segmentation. The science module uses a LIBS based spectrophotometer model to capture spectral information and analyze elemental composition of the geological samples. For a set of liquid and solid samples, the module was able to correctly match 60-65% of wavelength information from the sample spectra to identify basic elements (e.g. C, O, N, Fe). Finally, two and three qubit quantum-circuits were tested through IBMQ computer, which showed classical equivalent of 0 and 1 being generated through uncertain probabilities. While the experiment proved the concept of 'true' randomness, it cannot be determined to what scale quantum computers will aid in scientific data analysis of space exploration. In conclusion, the model was successful in demonstrating the basic scientific objectives of a Mars exploration rover.

Chapter 1

INTRODUCTION

1.1. Overture

Over the last few decades, ‘the space race’ has given rise to a billion-dollar industry, ushering in a new era for space science, robotics and exploration. The projects undertaken as a part of this initiative to explore planets beyond our own, brings together representatives from diverse academic and technical disciplines. Today, scientists and engineers work together to come up with the most cutting edge and efficient technological means to explore extraterrestrial planets and conduct scientific studies on them. The engineers bear the design load according to the specifications set by the scientists, in order to enable proper reception and representation of data from these planets. The main purpose for this is always tied to the age-old question that humanity has been chasing for the last few centuries – “Is there life beyond Earth?” and “Are there other planets which are habitable for humans one day?” The space revolution began right when humanity started getting familiar with the effects of the crucial modern-day problems, such as, depletion of natural resources, rising populations, increased emission of greenhouse gases and global warming etc. Missions to space now not only target the satisfaction of human curiosity but also hope to achieve key answers to solve these problems here on Earth.

1.2. Historical Background

One of the biggest challenges of space exploration missions on extraterrestrial planets is that most of the data collected requires autonomous management. This is the case for all satellites and exploration rovers from past and present that have been tasked with learning as much about the planets in our solar system. All the Mars expeditions till now have successfully demonstrated the increasing autonomous capabilities of the rovers currently traversing the Martian terrains in pursuit of scientific discoveries. The key objectives of these rovers are to be the ‘eyes and hands’ of planetary scientists to research and analyze the habitability potential of extraterrestrial planets. Onboard the rovers are scientific instruments capable of carrying out in-situ tests to determine signs of extinct and extant life.

However, for the scientific team relying on the data sent from these planets would have to wait an entire day or longer to be able to get the images from the rover's navigation camera, whereas the rovers are not designed to stop or go back if geologists or astrobiologists deem an area to be of scientific interest. Since any sort of transmission from, for example, Mars takes over a day to reach Earth, the updated versions of the Mars exploration rovers, such as, Opportunity, Spirit and Curiosity were designed with as much autonomy as possible, requiring minimum human assistance for decision making. The rover needs to make calculated assessments, without any delay or human intervention, about what actions to perform with regards to its surroundings in order to traverse foreign and unexpected terrains of a new planet, self-preservation and when to deploy the expensive science equipment for meeting its science goals.

1.2.1. Earlier Research

This project incorporates ideas inspired by the Autonomous Exploration for Gathering Increased Science (AEGIS) system [1,2,3], which specializes in acquiring specific spatial targets to deploy the hardware scientific sensors at, and the Onboard Autonomous Science Investigation System (OASIS) system, currently used on the Mars Exploration Rovers, with the objective of identifying unique science opportunities.

This project also closely observes the Mars missions which conducted soil science, terrain and rock-based analyses, such as, the Viking 1&2 (1972), Mars Pathfinder (1997) and Curiosity (2013). Key objectives of most of these rovers were to detect the presence of water and organic compounds. Moreover, the sensors for soil and rock analysis were paired with the specialized cameras on board to identify where the geological samples originated from. Most often, the basic markers for presence of water can be identified through the shape of the terrain. For example, distinguishing river channels or dried up river-beds can isolate areas of interest for exploration. This is where GIS (Geographic Information Science) tools were used to construct visual and spectral images of the surface of Mars to help identify possible landing and trekking sites. Spacecrafts and satellites, such as the Mars Reconnaissance Orbiter and the Mars Odyssey were equipped with instruments like Thermal Emission Imaging System (THEMIS) which helped identify the basic elemental distribution and geomorphic patterns on the surface of Mars. From there on, the exploration rovers conducted in-depth spectral analysis from the samples present in the region. The project will closely monitor the scientific data in accordance to these studies to ensure the output and conclusions derived maintain an acceptable standard of accuracy.

1.2.2. State of the art technology

The AEGIS software from [1,2,3], in development for over the last decade, has already been deployed to maintain the autonomous features for the Mars exploration rovers Opportunity, Spirit and Curiosity. The system accumulates a combination of different computer vision capabilities to determine areas of scientific interest and make decisions about deploying the very expensive on-board sensors. OASIS from [2,3,4] demonstrates more capabilities and features with navigation and path planning intelligence.

These technologies are some of the most advanced, among many others, in the field of artificial intelligence, computer vision and in-situ scientific experimentation. The unique challenges of space exploration provide the opportunity to introduce new ways of how we can approach engineering and technology to guide humanity to the next level of evolution. This project aims to explore these new domains, test out technological concepts that are being tested out on different planets, learn about interplanetary communication methods and the computational resources and approach towards understanding extraterrestrial data.

The vision for this project is not only to comprehend the scale of revolutionary engineering processes that go into space exploration rovers but to also replicate the ‘eyes and brains’ that make up the intellect of an artificial interplanetary scientist.

1.3. Objectives of the project

This project aims to build a model for a system capable of moving and traversing with respect to its (static) environment, which can distinguish different natural scenes to a certain degree, identify geological targets of interest, such as, rocks, and be able to filter out and prioritize a target, at which it can take sensor readings from. The sensors used should be relevant to the science objectives dedicated to determine key features about the geomorphologic history of the planet. The standard methods of identifying these features are through basic elemental composition study of rocks, soil and other geological sites of interest. This project will focus on breaking down the objectives and analyze the different methods that can be used to meet the base requirements for this project, along with testing procedures which will be conducted to make the system more accurate and robust.

1.3.1. Primary Objectives

- i) To replicate the basic mechanism of how Mars Exploration rovers operated as an observational planetary scientist.
- ii) To demonstrate a computer vision system for a rover, specifically designed for prioritizing scientific areas of interest in extraterrestrial terrains.
- iii) To deploy a scientific sensor in order to procure a basic estimation of the geological value of the terrain, to fulfill the standard scientific objective of detecting potentials for extinct or extant life.

1.3.2. Secondary Objectives

One of the key objectives of this project is to experiment with certain aspects of computational processing with quantum computers. This aspect is intended as a comparative study between the advantage of emerging quantum computing technology and classical computers, in terms of scientific studies related to space exploration. While the slow technological advancement in Bangladesh cannot directly contribute to space exploration within the next decade, this phase of the project aims to comprehend the complexities and effectiveness of quantum mechanics at the computational level for ambitious space expeditions.

1.4. Comparison with traditional methods

Exploration rovers are generally of interest in terms of planetary expeditions. There are very few instances where rovers have been used for any context in the environment of Earth. Traditional robotics engineering generally caters towards the different uses some fields might need from a rover-based model. There have been many attempts to recreate space exploration rovers by student teams to replicate the functionality of a complete Mars Rover but none of these attempts were reported to have been practically applicable for traversing terrains on Earth, let alone introduce helpful technology for extraterrestrial terrain traversal. This project does not attempt to recreate a complete rover system, but it only aims to explore the ‘scientific brain’ an exploration rover must have. One advantage of this research is that it has been developed through feedback from geologists working actively in the field. The Geological Survey of Bangladesh (GSB) collaborated through defining the challenges of collecting mass amounts of cohesive data through sending groups of geologists to remote sites. Currently, the system for geologists to collect data from the field is inefficient and time consuming, especially in the context of Bangladesh where the experts have to rely on expensive machines that cannot be moved from the lab.

This technology could have a scope for geologists on Earth to better increase frequency of observational data collected within a small timeframe and potentially reduce human inefficiency.

1.5. Impact of project on society

(Survey Questionnaire provided in Appendix A)

The context of this project emphasizes inquiries into some of the most relevant concerns for the future of human beings today. Addressing these concerns, a set of 50 participants, the majority of whom are students within the age group of 20-25 and working professionals of age 26-30, gave their opinion on the following issues:

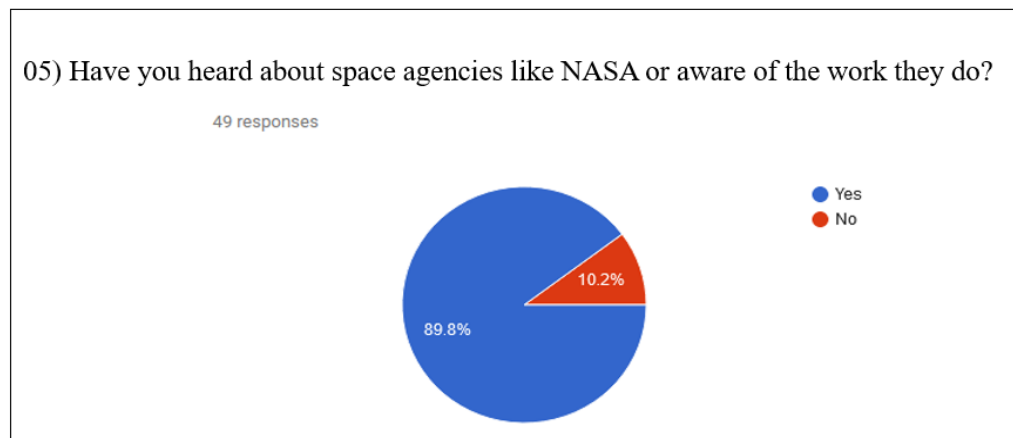


Figure 1.5.1: Overview of responses for question 5

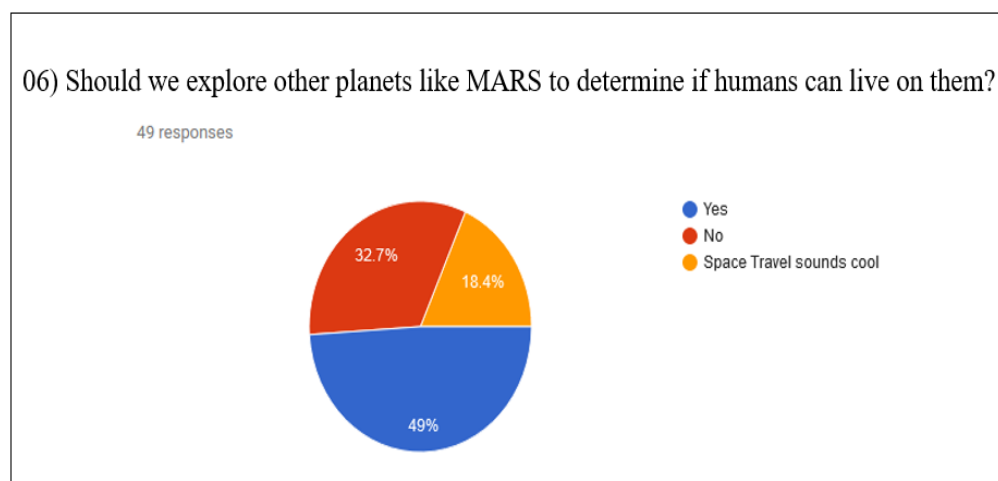


Figure 1.5.2: Overview of responses for question 6

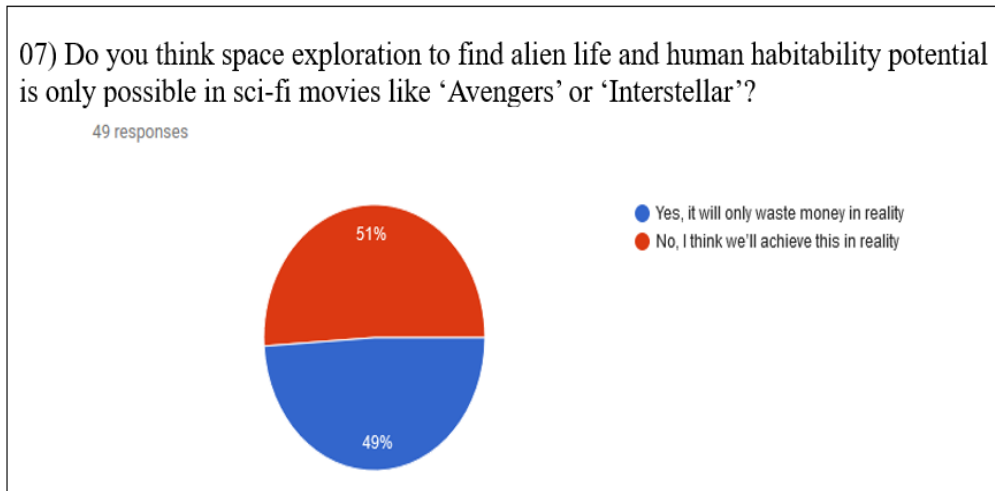


Figure 1.5.3: Overview of responses for question 7

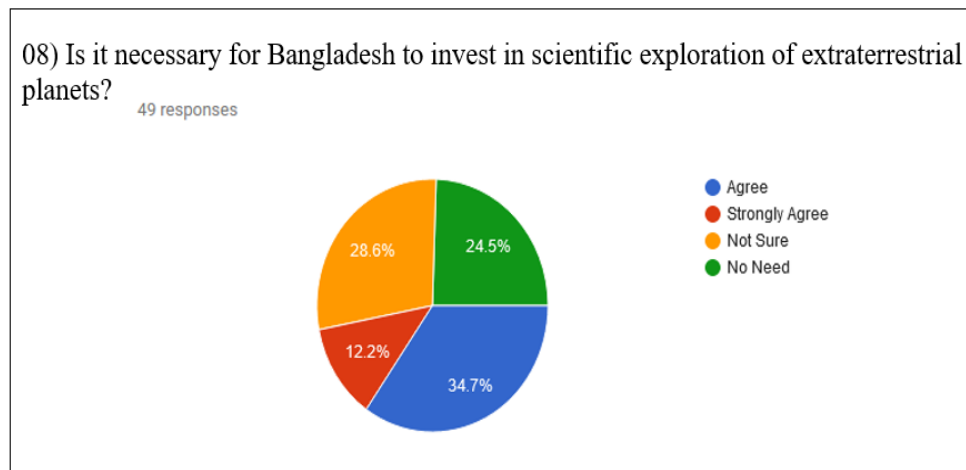


Figure 1.5.4: Overview of responses for question 8

The objective of this survey was to understand the impact of the project through how much people think this research is important. The concept of extraterrestrial exploration is relatively unfamiliar for the majority of the population in Bangladesh.

The small percentage with the appropriate educated base who are aware of these expeditions have been largely represented in the survey. The popular opinion, according to the results, is that space exploration technology brings hope for the survival of humans in the future, while an even margin does debate the worth economically.

1.6. Project Management

Table 1.6.1: The management for the tasks and milestones for the project

Tasks	Milestones
Problem Addressing	Proposal Submission
Researching for Solutions	Progress Report & Defense
Survey Questions	Project Book Completion
Readjustment of selected solution based on survey results	Poster & Summary Submission
Implement Design and Deployments	Pre-Defense
Testing and Results Analysis	Progress Defense
Critical Design Review	Final Defense

	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		Problem Addressing	131 days	Thu 6/13/19	Thu 12/12/19	
2		Data Analysis	23 days	Mon 9/30/19	Wed 10/30/19	
3		Data Analysis and Research at GSB	120 days	Sun 6/30/19	Thu 12/12/19	
4		Researching for Solutions	131 days	Thu 6/13/19	Thu 12/12/19	
5		Hardware Simulation	11 days	Mon 8/12/19	Sun 8/25/19	
6		Hardware Implementation	30 days	Tue 8/20/19	Mon 9/30/19	
7		Literature Review	38 days	Thu 6/13/19	Mon 8/5/19	
8		Implemented Designs and Deployments	91 days	Thu 8/8/19	Thu 12/12/19	
9		Sensor Integration	22 days	Fri 8/30/19	Mon 9/30/19	
10		UI Development	34 days	Thu 8/8/19	Tue 9/24/19	
11		Vision Development	28 days	Thu 8/8/19	Sun 9/15/19	
12		Testing and Results Analysis	54 days	Mon 9/30/19	Thu 12/12/19	
13		Testing and Debugging Phase	24 days	Sun 10/20/19	Wed 11/20/19	
14		Quantum Simulator Testing	23 days	Mon 9/30/19	Wed 10/30/19	
15		Critical Design Review	43 days	Tue 10/15/19	Thu 12/12/19	
16		Book Preparation	33 days	Tue 10/15/19	Thu 11/28/19	
17		Wrap Up for Final Defense	18 days	Tue 11/19/19	Thu 12/12/19	

Figure 1.6.2: The Gantt Chart for The Project Management (1)

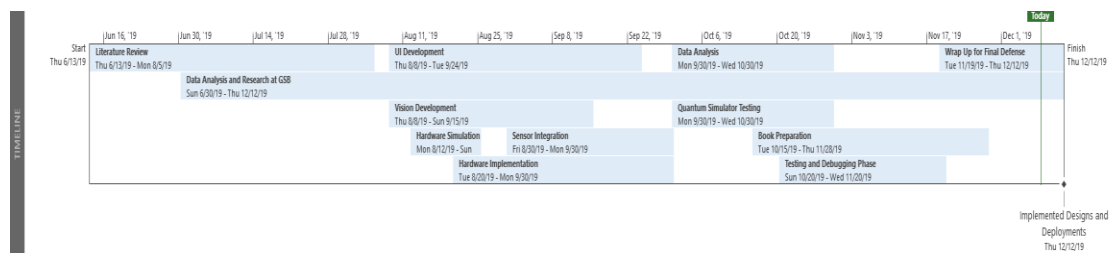


Figure 1.6.3: The Gantt Chart for The Project Management (2)

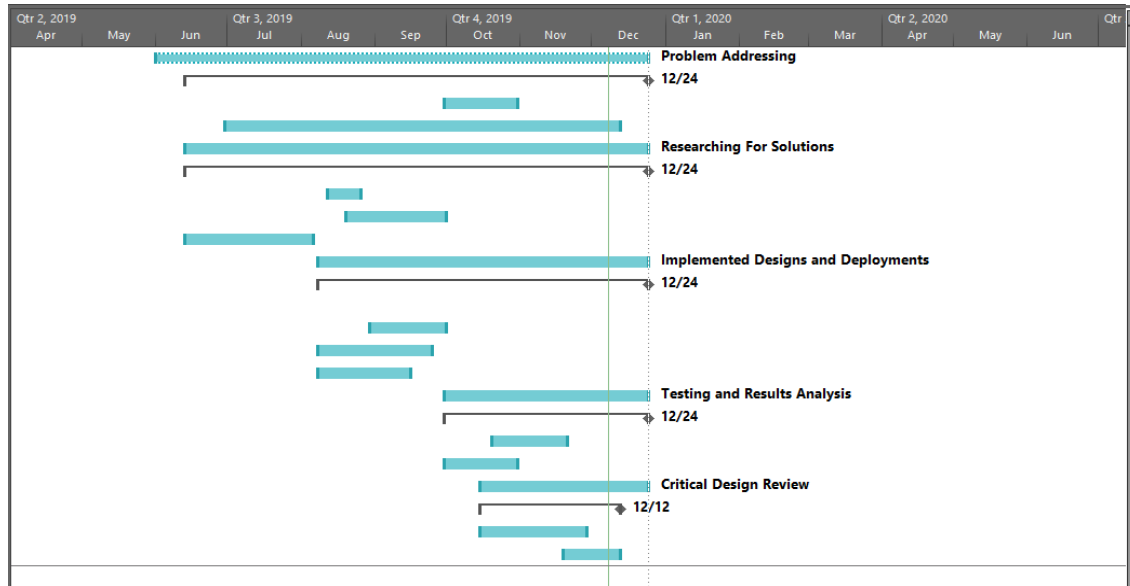


Figure 1.6.4: The Gant Chart for The Project Management (3)

1.7. Organization of book chapters

Chapter-2: Literature Review with in-depth investigation

Chapter-3: Image Processing with Segmentation and Detection

Chapter-4: Instrument for Analyzing Geological Samples of Interest: Spectrophotometer

Chapter-5: Complete prototype design and deployment

Chapter-6: Experimental Analysis with Quantum Computing

Chapter-7: Conclusion

Chapter 2

LITERATURE REVIEW WITH IN-DEPTH INVESTIGATION

2.1. Previous work

Basis for study:

This chapter investigates into the key concept that goes into designing an autonomous field scientist. This required for the team to assess how a scientist working in the field may approach the task. The following figure illustrates the basic block diagram of what the project aims to represent the core of an exploring “scientist” rover.

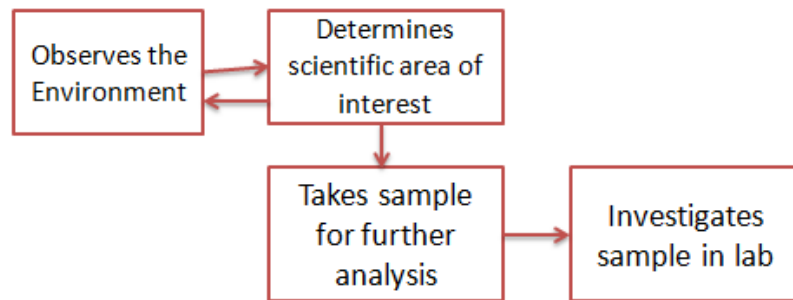


Figure 2.1.1: Basis of the study

The sections of the chapter therefore investigate research into the domains of

- Building custom computational system for developing rover ‘brain’
- Computer Vision for geological scene sampling
- Scientific equipment for analyzing geological sample of interest

Lastly, this section also studies some advantages of using quantum computing for all these domains in the context of space exploration.

2.2. Studying the Brain of Exploration Rovers

2.2.1. Background Study for Custom Operating system of the Raspberry Pi

To design the custom operating system of the Raspberry Pi, significant background study was conducted to evaluate which distributions could deliver the required features. Considering the hardware provided for Raspberry Pi 3, Model B comes with the 64-bit quad-core processor, aboard connectivity, and Bluetooth access and USB options. Now fast forward to the design approach as the mechanism is formulated for a remote sensing connectivity which must be viewed for its associated devices which can come in the form of Laptops, Desktops, Ipads, Tablets, Phones or any displayable device. In order to achieve this aspect, it is very much simple and easy with the integration of some simple sensory like a Bluetooth Shield module as well as for Wi-Fi and with a router for any sort of Ethernet port connectivity. Also, a wiring will take place which will result in higher portion of power consumption. Taking into account pricing and combining will also bring out a larger bounded design in the scope for the project. So, there are many issues to be resolved in respect to the context as the design must be precise and optimal when performing the operations.

This is where our design approach kicks in and plays an unorthodox role at pursuing the formulation. Using coding mechanisms and patching of the distributions it can be optimized to a design which will work as per our modification and provide output along with feedback in the best way possible. The OS that is basically implemented in the provided RPi is a hybrid of the available distributions and is capable of performing at an artificial level of integrity as it has its own preferences for performing the operations associated with it. Also, it will optimize itself based on the sensory integrated to it and perform accordingly based on the architecture it is included and used upon. As for the advantages of this setup and composition, firstly, no additional hardware or sensory will be required for the Wi-Fi activity and Bluetooth connectivity along with the Camera Navigation.

2.2.2. Video feedback and Image processing

Given the scopes, the system OS will work in a logical reasoning manner which is what a basic artificial system would implement and compute on. For the modulation of the hardware implementation, the power proficiency has to be viewed and noted as well. As from a traditional aspect, it would require different modulation whereas now it can all be performed within the 5V range and no additional sensory is required either for the functionalities resolving in Remote Connections and Connectivity.

Also, with the addition of a Smartphone camera, the Navigation aspect can also become a more-simple run down with respect to the design schematics. An Internet protocol camera, or information science camera, maybe a form of digital video camera unremarkably used for vision work, and which, in contrast to analog television system (CCTV) cameras, will send and receive knowledge via a network and therefore the net.

2.2.3. Background Study for the Computer Vision work

Digital image processing is termed under the fact, that the use of computer vision algorithms to perform the image process on digital pictures. As a subcategory or field of digital signal processors, the digital image process has several benefits over analog image processing. With the modernized system infrastructures, it has become rapidly developed to a level of supremacy. However, depending on scenarios presented and situations it has a decay effect. [10] For instance, today's concepts like Artificial Intelligence, Machine Learning, Deep Learning, Data Binding, Manipulation, Extraction, Mining, Convolutional Networking are some of the perks to push the boundaries to a new height of efficiency. Under this aspect, many techniques have been deduced and approached to acquire desired results from entry inputs to satisfy the needs of computing capability [12-22].

Like the Curiosity rover for MARS Exploration or the OASIS system used for capturing and gathering of data of extraterrestrial planets imaging plays an impacting role. As it is said like the verse, a picture tells of a thousand stories. The same prospect is true for the project deployment undergone for this research and conduction of the applied techniques [15-25].

2.2.4. Image Detection

Image recognition refers to technologies that determine places, logos, people, objects, buildings, alternative dimensions and several other variables in pictures. Pictures carries information within the variety of 2-dimensional matrices. In fact, image recognition is classifying information into one class out of the many. One common and a very important example is optical character recognition (OCR). OCR converts pictures of typewritten or written text into machine-encoded text. The major steps in image recognition method consists of gathering and organizing information, build a prophetic model and use it to acknowledge pictures.



Figure 2.2.4.1: A simple illustration of an Image Detection process

The human eye perceives a picture as a collection of signals which are processed by the cortical area within the brain. This ends up in a vivid expertise of a scene, related to ideas and objects recorded in one's memory. Image recognition tries to mimic this method as like the Figure 2.2.4.1. The primary step in image classification is to change the image by extracting vital data and deed out the remainder. As an example, within selected images if one wishes to extract information from the background one will notice a major variation in RGB pixel values.

However, by running a footing detector on the image we will change it. The model generated still simply recognize the circular form of the face and eyes in those edge pictures then one will conclude that edge detection retains the essential data whereas discard non-essential data. Some well-known feature descriptor techniques are Haar-like options introduced by Viola and Jones, bar graph of familiarized Gradients (HOG), Scale-Invariant Feature remodel (SIFT), accelerated strong Feature (SURF) etc. with Mobile SSD Net_V2, Faster RCNN, COCO_Geometry_V3, ENet.

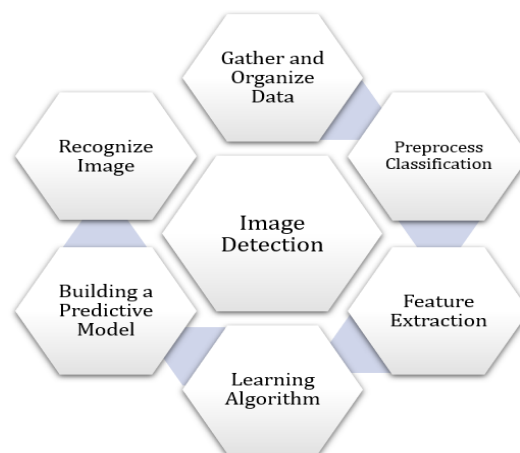


Figure 2.2.4.2: A Flowchart for Image Detection

2.2.5. Image Segmentation

Segments represent objects or components of objects, and comprise sets of pixels, or “super-pixels”. Deep learning will learn patterns in visual inputs so as to predict object categories that form up a picture. The most captive deep learning design used for image process may be a Convolutional Neural Network (CNN), or specific CNN frameworks like AlexNet, VGG, Inception, and ResNet. models of deep learning for computer vision square measure usually trained and dead-on specialized graphics process units (GPUs) to scale back computation time. Image segmentation helps verify the relations between objects, additionally because the context of objects in a picture. Applications embody face recognition, variety plate identification, and satellite image analysis. Industries like retail and fashion use image segmentation, as an example, in image-based searches. Autonomous vehicles use it to grasp their surroundings. The perfect examples for this particular case are the refined MARS rovers for scientific exploration.

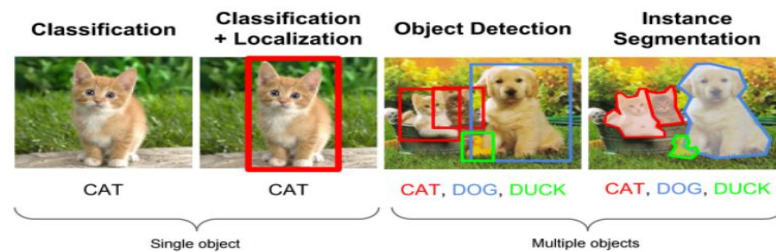


Figure 2.2.5.1: A simple illustration of an image segmentation process

The powerful depictive ability of ResNet boosts laptop vision applications like object detection and face recognition. **Atrous abstraction pyramid pooling (ASPP)**—provides multi-scale data which is illustrated in the Figure 2.2.5.2. It uses a collection of atrous convolutions with varied dilation rates to capture long-range context. ASPP conjointly uses international average pooling (GAP) to include image-level options and add international context data.

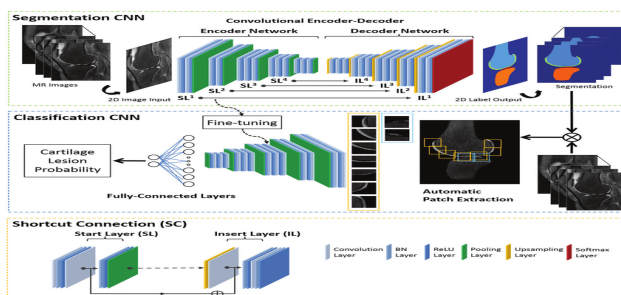


Figure 2.2.5.2: A simple illustration of an image segmentation process

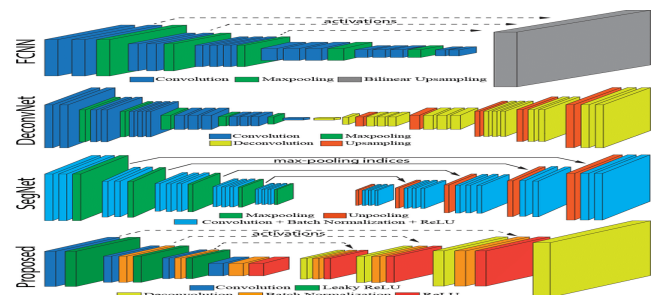


Figure 2.2.5.3: An overview schematic of the processing involved

Convolutional Neural Networks (CNNs) image segmentation with CNN involves feeding segments of a picture as input to a convolutional neural network, that labels the pixels. **Fully Convolutional Networks (FCNs)** Ancient CNNs have fully-connected layers, that can't manage completely different input sizes. FCNs use convolutional layers to method varied input sizes and might work quicker. **ResNet**—a deep convolutional network (DCNN) from Microsoft. It provides a framework that permits coaching thousands of layers whereas maintaining performance illustrated in Figure 2.2.5.3.

From this pipeline staging the images that are finalized for prediction turns in associated layers that form the final segmented image that is illustrated within the Figure 2.2.5.4.

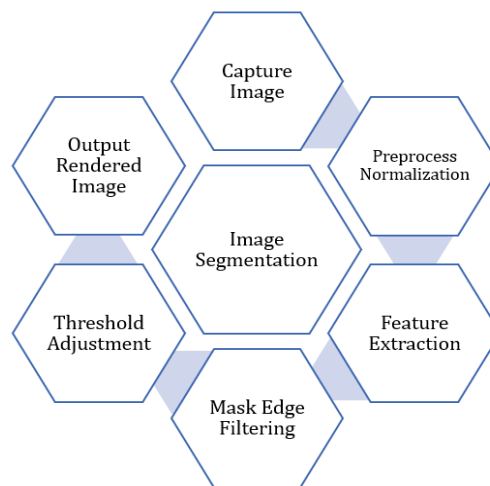


Figure 2.2.5.4: A Flowchart for Image Segmentation

2.3. Scientific Instruments deployed on Mars Rovers

Understanding the geochemical processes of extraterrestrial planets has been a crucial part of space exploration missions. It all comes down to the basic questions about the planet's evolution, all the way towards the habitability potential of the planet through past, present and future. This study is generally based on what we learn from Earth and its evolution processes which enabled life of all forms to sustain on its surface. Till date, since Mars has been the only planet on which rovers have been able to successfully conduct scientific studies for a prolonged period of time, it is worth studying the array of scientific instruments that have been deployed along with them. Over the course of years, there have been many variations of spectrometry-based equipment deployed on the Mars rovers.

The model upon which the spectrometer for this project would be designed came from the instrument already deployed on the Curiosity rover called ChemCam. Short for ‘The Chemistry and Camera tool’, ChemCam is a combination of laser, camera and spectrograph to identify the chemical and mineral composition of rocks and soil. The basis for this type of spectroscopy is called LIBS (Laser-Induced Breakdown Spectroscopy), which is a powerful tool used for remote geological sensing purposes. The principle of LIBS lies in firing a powerful laser repeatedly, concentrating on a point on the surface of the sample under inspection, to produce a resulting plasma from the dominant elements and minerals of the sample. The spectral information is then picked up by a camera which breaks down the optical information to be displayed as a graph. Figure 2.3.1 illustrates how the laser is fired at sample.

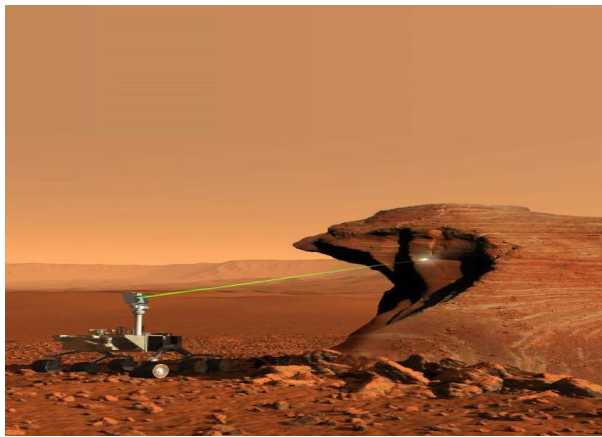


Figure 2.3.1: The Laser-Induced Remote Sensing for Chemistry and Micro-Imaging instrument will identify atomic elements in martian rocks. Credit: NASA/JPL-Caltech/LANL/J.-L. Lacour, CEA

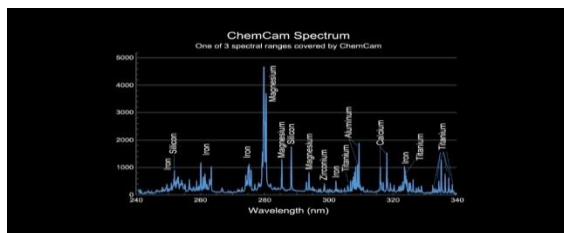


Figure 2.3.3: **Example of a Spectrum from Curiosity's ChemCam Instrument:** This image provides an example of the type of data collected by the Chemistry and Camera (ChemCam) instrument on the Mars Science Laboratory mission's the Curiosity rover. Credit: NASA/JPL-Caltech/LANL

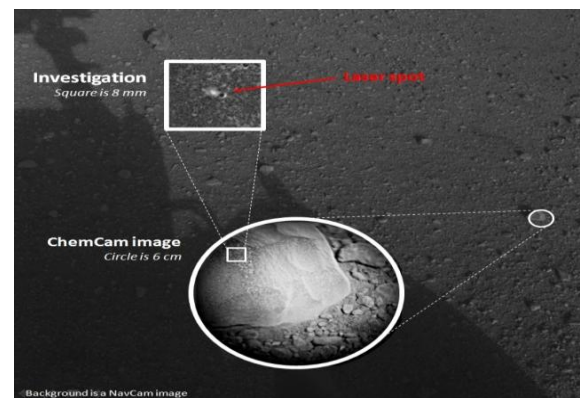


Figure 2.3.2: **First Laser-Zapped Rock on Mars** which is composite image, with magnified insets, depicts the first laser test by the Chemistry and Camera, or ChemCam, instrument aboard NASA's Curiosity Mars rover

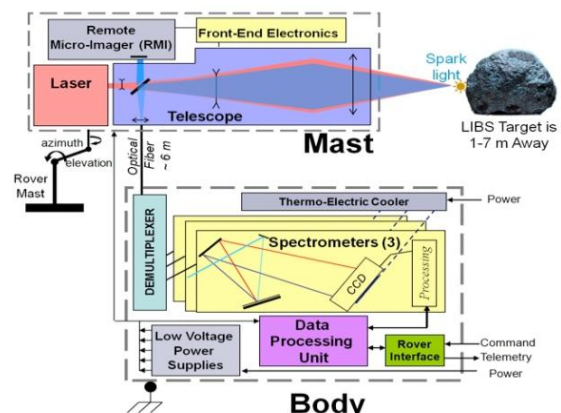


Figure 2.3.4: The figure shows a functional block diagram of the ChemCam suite

2.4. A brief study into quantum computing

Over the last few decades, there has been much hype over the concept of quantum computers. Even though universal quantum computers are still years away from becoming a reality, there are still powerful supercomputing devices today which are built on quantum mechanical models. These devices have become a crucial part of space exploration missions, among other fields of use. NASA space missions are currently using a quantum annealing computation technology that aid in investigating areas where quantum algorithms might someday dramatically improve the agency's ability to solve difficult optimization problems in aeronautics, Earth and space sciences, and space exploration. Applications relevant to NASA include machine learning, pattern recognition, mission planning and scheduling, distributed navigation and coordination, and system diagnostics and anomaly detection. This technology is a 2,048-qubit D-Wave 2000Q quantum computer hosted in the NASA's Quantum Artificial Intelligence Laboratory (QuAIL). The QuAIL project is a collaborative effort among NASA, Google, and Universities Space Research Association (USRA) to explore the potential for quantum computers to tackle optimization problems that are difficult or impossible for traditional supercomputers to handle.

Industries are now looking towards leading quantum computing research facilities at Google, Microsoft and IBM, who are currently leading the billion-dollar race in this field. However, it is still hard for the world to truly comprehend what advantage these quantum computers can provide over the well-established classical computers. The main reason is the mathematics involved in quantum programming are only being developed and tested, while most engineers would find the concepts complex. This is due to the nature of quantum mechanics itself. Classical computers carry out logical operations using defined and static physical states. These are usually binary bits 1 or 0, meaning its operations are based on one of two states. In quantum computing, operations use the quantum phenomenon to produce what is known as a Qubit. These states are the undefined properties of a quantum level object before they have been detected, or more accurately, measured, such as the spin of an electron or the polarization of a photon. Rather than having a defined state, unmeasured quantum states occur in a mixed 'superposition' state (Figure 2.4.1). Other quantum mechanical phenomenon includes entanglement where changes made in the property of one qubit can affect another 'entangled' qubit irrespective of distance or barriers (Figure 2.4.2). Building a functional quantum computer requires holding an object in a superposition state long enough to carry out various processes on them. Unfortunately, once a superposition meets with materials that are part of a measured system, it loses its in-between state in what's known as decoherence and becomes a classical bit.

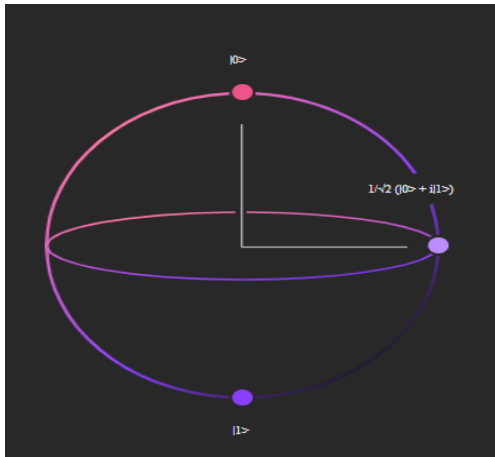


Figure 2.4.1: Superposition

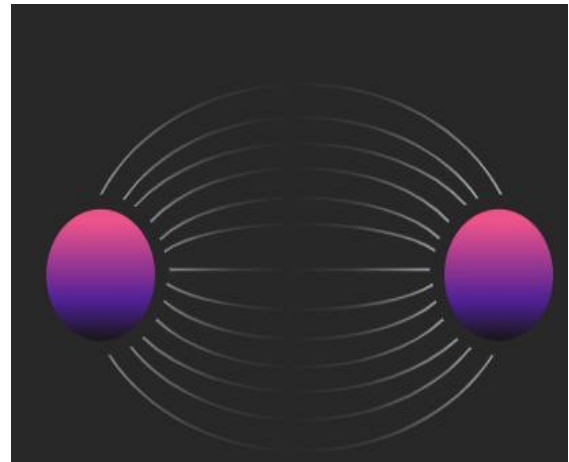


Figure 2.4.2: Entanglement

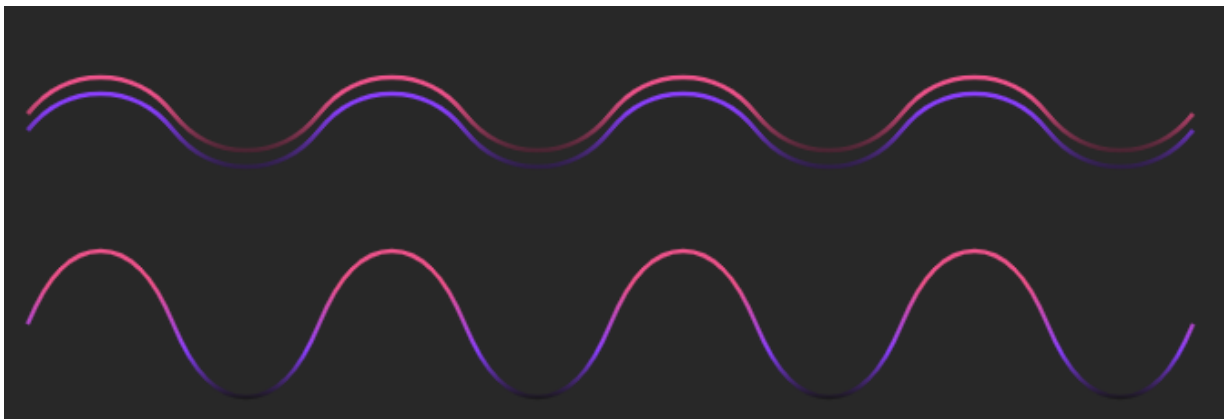


Figure 2.4.3: Interference

Chapter 3

Image Processing with Segmentation and Detection

3.1. Introduction

This chapter discusses the software development with the computation processing to achieve the desired goals set for the project. The processing performed is categorized in a sequential manner. Firstly, the images captured for the point of interest with identification is detected using electronical peripheral devices which are described in the further sections of this chapter. Secondly, the detected image is rendered and segmented using various algorithms for the image processing pipeline required for the project. Lastly, the customized processing pipeline illustrated in Figure 3.1.2 is comprised solely of python scripts.

The final output and visualization are further discussed in this chapter, highlighting proficiencies and variations from the concurrent techniques used generally. Along with that, this chapter also includes the software distribution used for conducting the operations in order to achieve the desired output. The organization of the entire project model is outlined in the Figure 3.1.1.

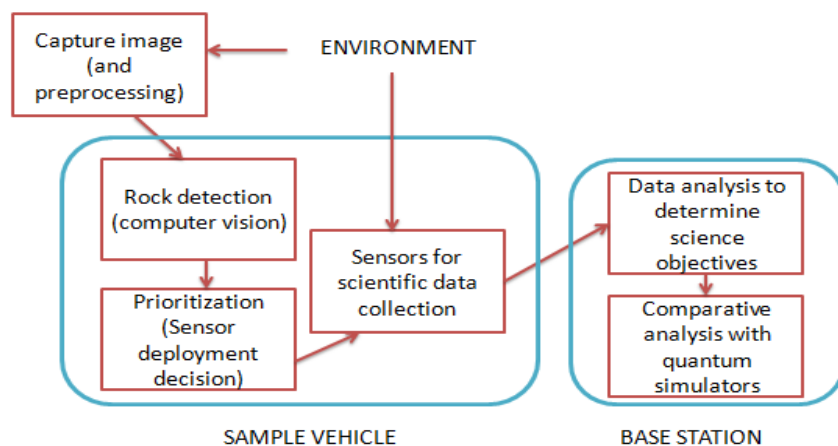


Figure 3.1.1: A workflow of the custom designed system

3.2. Methodology and Modeling

This section provides insight to the experiments performed with their associated trials and errors for gathering the desired results. For instance, custom detection and segmentation of images in real-time is still under development. The concept and computation require a hierarchical order in terms of the python scripting schemes, with different sets of algorithms for the generation of custom models, including image classification and labeling of the segmented datasets used for the processing. Device peripherals utilized for achieving the goal consists of a Raspberry Pi 4 model B and the Coral USB accelerator.

The sequence of objectives at different stages is broken down in Figure 3.1.2 which is provided below with its illustrations.

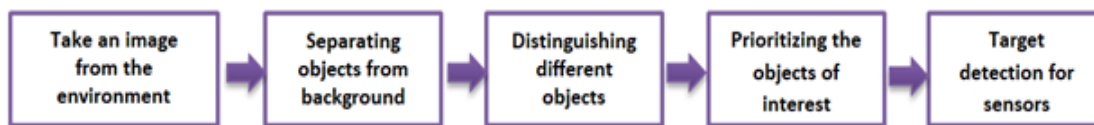


Figure 3.1.2: Image processing stages

Each of the stages from Figure 3.1.2 focuses on identifying just a single rock from the scenic image. Due to the fact rocks contain significant traces of history of the evolutionary and geomorphological processes of a planet, advanced computer vision techniques focus on features such as shape, size, texture and even on some occasions color. These features can characterize and categorize the geological formations of the rock and where it possibly originated from, giving indications of past existence of water on the planet. Since all life forms on Earth are biologically reliant on water, rocks which indicate past traces of water bodies prove to be samples of scientific interest. Afterwards, a target on rocks, such as veins or unique texture or color spots, is acquired and high frequency laser-based sensors are used to study and analyze its elemental nature for presence of hydrogen, oxygen and, most importantly, hydrocarbons or organic compounds. The vision work done is compartmentalized in the following numbered sections with its illustrations.

1. **Analyzing image while the rover is moving.** The initial development will focus on working with a single frame of the scene encountered. Natural scene of the environment on Mars is dominantly barren, therefore the images and the algorithms that will be trained will focus on distinguishing between background (sky, clouds, horizon) and objects on the ground.

Other formations similar to rocks are taken into account but would not be considered rocks of scientific interest therefore the accuracy of identifying a standard rock should be ensured within 80-90%. A method used as part of OASIS system [1,2,3] for distinguishing sky from a grayscale image is by finding seeds to see if the sky is present, and if so, areas of low variance are identified and lower bounds of the sky area are delineated by edge detection to determine the horizon line [2,3,4]. Clouds and dust with distinctions are done to eliminate error in identifying closed edge formations with rock-like similarities. The system for this project will only focus on identifying rocks from ground area (eliminating sky).

2. **Rock detection using edge detection methods** to find closed contours. Few issues are present that need prior attention such as relative position or spatial orientation of the rock from the point of perception. Afterwards, image segmentation can be used to distinguish foreground from objects of interest by detecting characteristic patterns within pixels groups, and possible matching with a template. There are also methods for detecting rocks using their shadows but that will only be considered if the accuracy of the previous methods is not satisfactory. Systems deployed on the Mars rover uses image normalization before running it through an edge preserving a smoother filter and enhanced using unmask sharpening [2,3,4]. To further enhance the detection accuracy, denoising filters can be used to remove textures that might be misidentified as smaller rocks.

3. **Filtering and morphological clean-up** are used to increase the accuracy of rock detection. Filters can help eliminate false positives, which can arise from deep textures and stray pixels that are not found clustered along edge groups. Morphological clean-up could be used to close down broken and fragmented edges and form the closed contour. Out of all the rock detection systems compared in [3,4,5] the ‘Rockfinder’ system displayed the most accurate result by utilizing a multi scale image pyramid and combining the results at the end.

4. **Data set matching, prioritization and target acquiring** (coordinates) come into play only after multiple rocks have been detected and one rock has to be selected to deploy the scientific instruments at. This process will use pixel values to determine intensity (light or dark color distinction) of rocks, texture, shape (eccentricity and ellipse fitting) and size. These values will be matched with a dataset established prior to deploying the system to find the values corresponding to a high scientific potential of rocks.

However, our deployment is for rocks which is very hard to calibrate due to the inconsistent resolutions and attributes involved. Scaling from color, hue and saturation, the layers involved make a huge number of attribute selections both for prediction and detection. During the training and testing of the processing of rock classification and identification, a total number of 42 categories came about to classify and create the custom model for deployment. Along with that, a total number of 437 images consisted for training and testing phases. In order to generate these a few steps needs to be followed in a sequence. As a consequence, the desired approach for model creations consists of the following given below.

- Model selection on the choice of interest
- Features and properties to use the model
- Train the model using a specific pipeline
- Test and gather data with its associated records
- Optimize the results in accordance to the perspective deployment
- Evaluating results from the model
- Making predictions and illustrating its accuracy

From [11-22] it is evident that the stage of machine learning is initiated once the model is step by step optimized, or the model learns the dataset.

3.2.3. Design Formulation and Deployment of Models

The project uses the platforms of TensorFlow, OpenCV, Yolo, DarkNet, which resulted in attaining the expected results which are given within Figures 3.2.3.1-3.2.3.3 Firstly, TensorFlow which is a combination of many distribution packages works as a dedicated workflow model illustrated in the figure 3.2.3.1. Then comes OpenCV which is an advanced version from tensors where the processing is done between an architecture deployed environment. **OpenCV** A pair of libraries that made major changes to the C++ interface, aiming at easier, additional type-safe patterns, new functions, and higher implementations for existing ones in terms of performance (especially on multi-core systems). The mentioned platforms utilized in line with the process segmentation pipeline and the project experiment analysis outperformed on each of the models. Each perspective model accumulated under the API interface protocol with their generated TF records running under their respective kernels for the identification and classification.

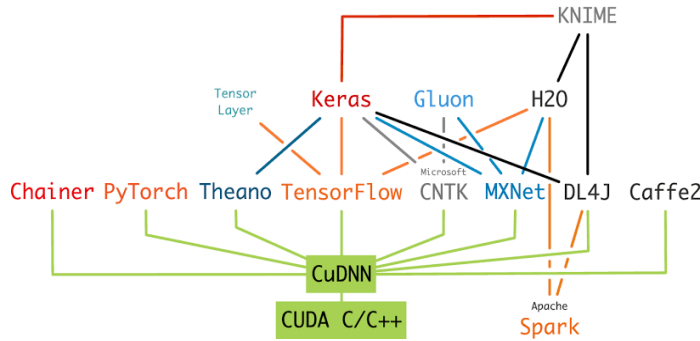


Figure 3.2.3.1: An overview of TensorFlow in action and deployment [7] [9]

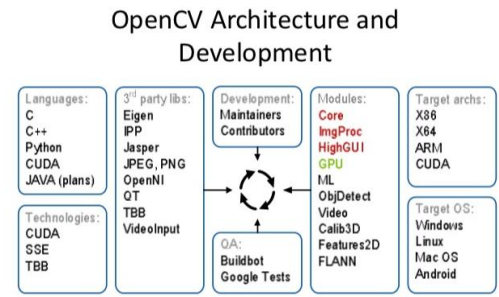


Figure 3.2.3.2: An illustration of OpenCV in action [21] [22]

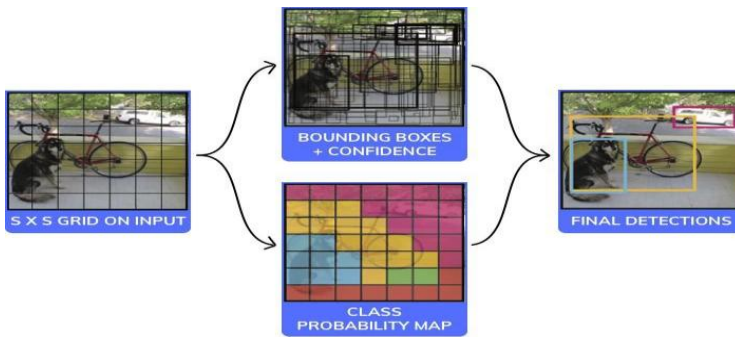


Figure 3.2.3.3: An illustration towards Yolo [24] [25]

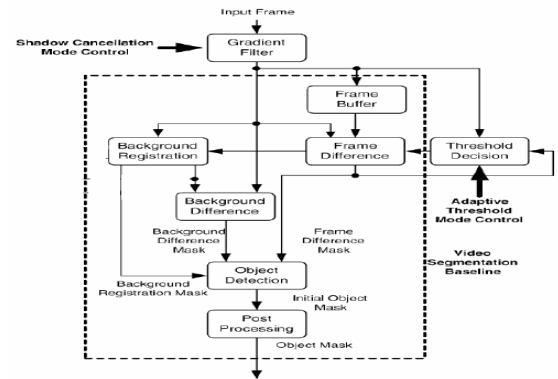


Figure 3.2.3.4: An illustration of the deployed formulation design

3.3. Finalized Models for Result Prediction

The same process has been applied to calculate the rate of probability and detection in different sections for an input image. Using the LabelImage tool, the layers are identified and using the values propagating for Alpha and Beta levels, each image generated, segmented sets of rocks shown in the output segmented image provided within the Result and Analysis section. For error tracing and better results in terms of graphical analysis mathematical models got introduced to get the optimum accuracy of each input corresponding to their designated outputs.

Apart from that, resolving for signal noise ratio reduction, mask edge filtering, size focus area, choice of interest anticipation with grayscale imaging in the RGB color space also outperformed to differentiate between consecutive output results for given input images illustrated in the Result Analysis Section with Text Labeling.

3.4. Result Analysis & Critical Design Review

3.4.1. Implemented Model Results

The processing and each illustration are labeled for its approached design formulation and deployment. Figure 3.4.1.1 illustrates the captured image given as input with corresponding grayscale space in an edge triggered environment. Perspective threshold is also provided in terms of its area and surrounding in reference to the natural scene. Figure 3.4.1.2 illustration is provided for the scope with respect to the objects detected (rocks) within the given natural scene that is real time buffer in a particular frame set. Then running the processing scheme its able identify a total of 296 objects in the frame labeled for our custom rock detection model. Furthermore, the mask edge filtering frags out the unwanted SNR values to generate an optimum output consisting of the objects segmented, detected and identified.

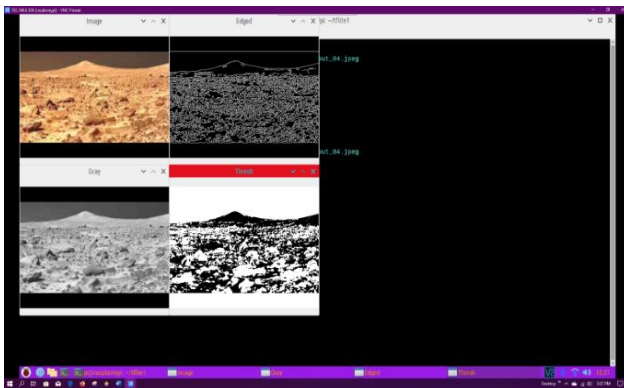


Figure 3.4.1.1: Image processing in real time feed

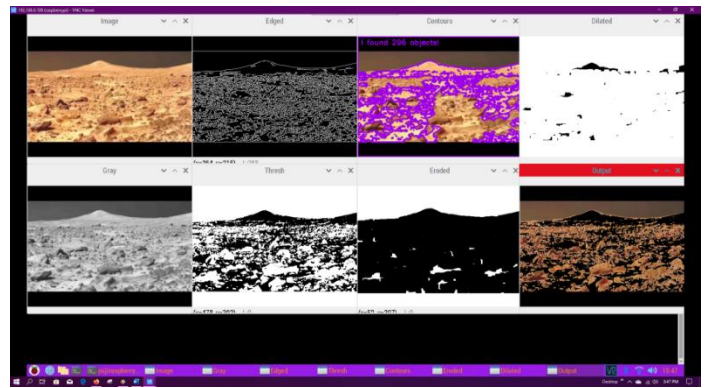


Figure 3.4.1.2: Output segmentation from the designated input image

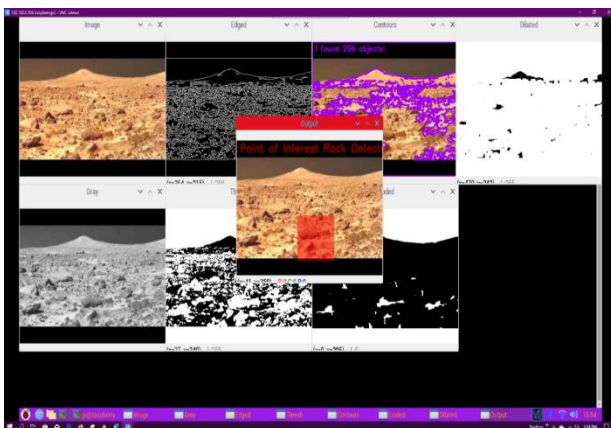


Figure 3.4.1.3: Point of interest detection

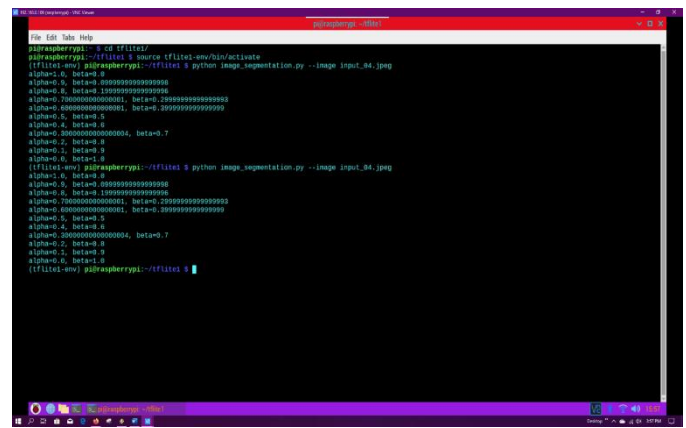


Figure 3.4.1.4: Image meta data generation

Figure 3.4.1.3 illustrates the buffer sequence of frames from the automated model, which recognizes and focuses onto a point of interest to examine. Based on the feed and frames and the models' generated alpha and beta values, the image processing starts to identify the geological interest point. Furthermore, from the roaming and video rendered from the feed the model will try to process the gathering of information it has trained itself towards and will deploy when it comes across that type rocks or environment. Since the deployed robot will be constantly taking video feed or frame buffers. Basically, the model will try to identify its scope of the interest point, then in accordance to the prediction the rover will roam to the area identified. The captured image will then be processes, which will be in line to resolute inside its layer of frames. Depending on calculations, the model tries to segment the best and optimum result possible for prediction and identification. Figure 3.4.1.4 illustrates the generated values processed for alpha and beta for the scoped input image rendered through the model.

3.4.2. Generated Prediction Results



Figure 3.4.2.1: Rock Classification

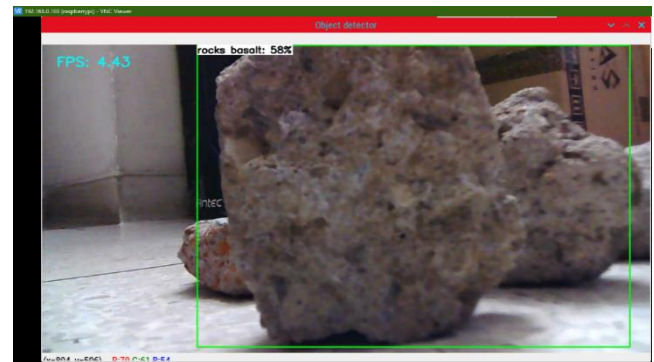


Figure 3.4.2.2: Rock Detection

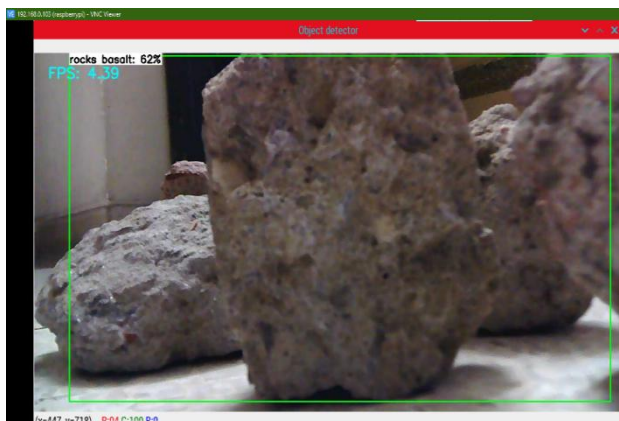


Figure 3.4.2.3: Rock Identification

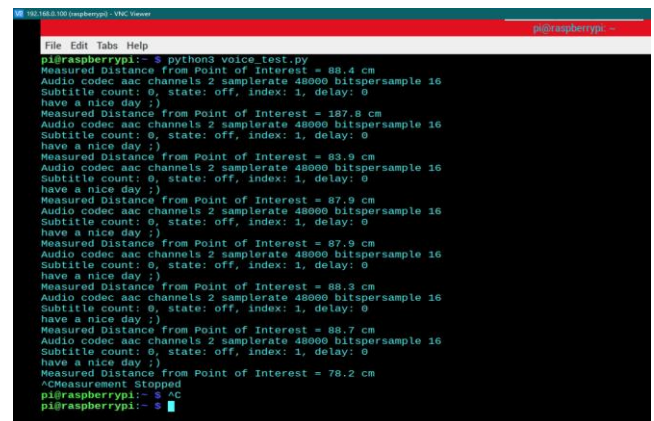


Figure 3.4.2.4: Point of Interest distance

The Figures 3.4.2.1, 3.4.2.2, 3.4.2.3, 3.4.2.4 illustrations are the real-time feedback and processing computation in terms with the customized model for the designed approach of the project. This framework has been optimized and categorized based on the initial basis of the study provided in Figure 2.1.1 and the probabilistic accuracy is also labeled with the context of the video feed. Also, the measurement distance calibrated and returned from the ultrasonic sensor is provided in terms of ('cm') as per accordance to the point of interest and field of view. When the desired aspect is found the measurement is terminated.

3.5. Design review/summary

Concerning the deployment mechanism, the processing is divided into two subsets of conductivity termed under detection and segmentation. Provided the scope of research it is evident that the initial goal for the results is met with a very good proficiency. Although for some aspects the experiments conducted did not produce the desired results. Based on the incompatibility and development of a new hybrid system architecture there are a few setbacks. The real time detection for given input images with natural scenes does meet with the requirement. Along with the segmentation, the system is capable of detecting the objects (rocks) at an efficiency of almost around 88.8%. Also, the mask edge filtering with the point of interest selection is illustrated with high alpha and beta level values based on rock formations. The video buffers for image classification also meets, but the prediction capability is seen a bit lower than average with slight decrease in its proficiency.

Chapter 4

Instrument for analyzing geological samples of interest: Spectrophotometer

4.1. Background

Considering the challenges of building a rover, which practically amounts to a robotic autonomous scientist on wheels, much geological sample analysis equipment, generally found in a large laboratory set-up on Earth, needs to be modified and scaled down for in-situ data collection. After identifying a geological area of scientific interest through machine vision, the objective of the “scientist” is to take a small sample from the source, prepare and process it to identify its molecular and elemental composition.

Challenges:

- In-situ scientific sample analysis with minimum impact on the environment
- Able to produce fast results since rovers need to manage time and resources to trek throughout miles to complete a day’s mission
- Being able to identify key scientific data that can contribute towards the investigation of life, water and habitability of the planet.
- Scientific instrument should not be resource intensive in terms of size, operability and on-board computing/processing since it is effectively sharing very limited hardware and software space within the rover

4.2. Methodology

Investigating further into the core concept of the kind of spectroscopy used for ChemCam, the crucial aspects are:

- The laser has to excite the sample to the point of vaporization. With enough laser emissions, subsequent plasma could form consisting of the liberated ions of the elements that make up the sample.

- The plasma light should be captured without much interference from the environment. To achieve as much accuracy in capturing the optical data with minimal noise, lenses are used to direct the captured light to an optical fiber.
- The captured light is broken down into its constituent wavelengths through a diffraction grating.
- The diffracted light is then fed to a light sensor or a powerful camera which can discern slight differences in wavelengths.
- A subsequent spectral graph on Intensity vs. Wavelength can then be plotted which serves as the basis for understanding what the samples are composed of.

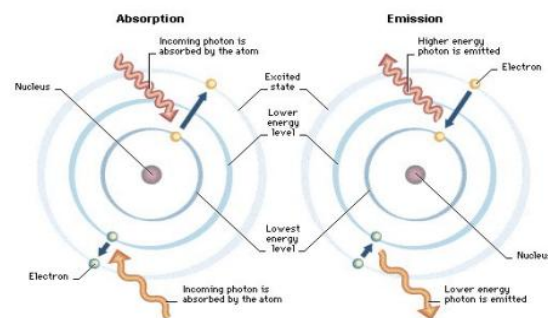
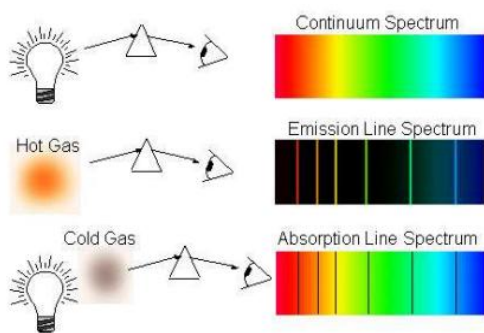


Figure 4.2.1: Examples of a continuous, an emission and an absorption spectrum with diagrams illustrating how the spectra were obtained.

Figure 4.2.2: Model of an atom showing absorption and emission of photons.

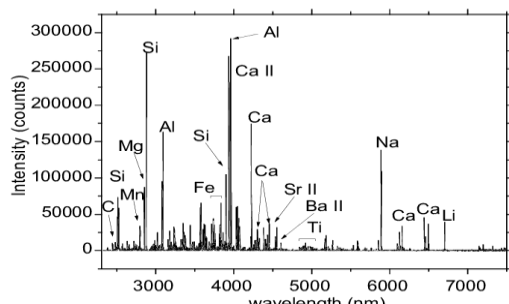


Figure 4.2.3: Spectra of rock sample (Andesite) from Mars

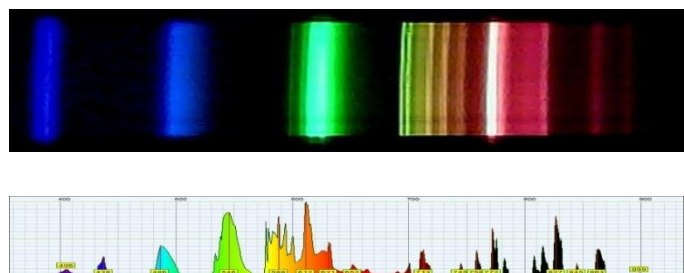


Figure 4.2.4: Image and Spectrum of a Fluorescence Lamp (mercury emission at **546 nm**, **436 nm** and **405 nm** and the doublet at **577 nm** and **579 nm**)

Considering all these factors, it can be deduced that for analyzing geological samples in-situ with minimum intrusion to the environment, a Laser-Induced Breakdown Spectrometer model can be incorporated into the design of the proposed scientific instrument. Basic model of LIBS setup is illustrated in Figure 4.2.5.

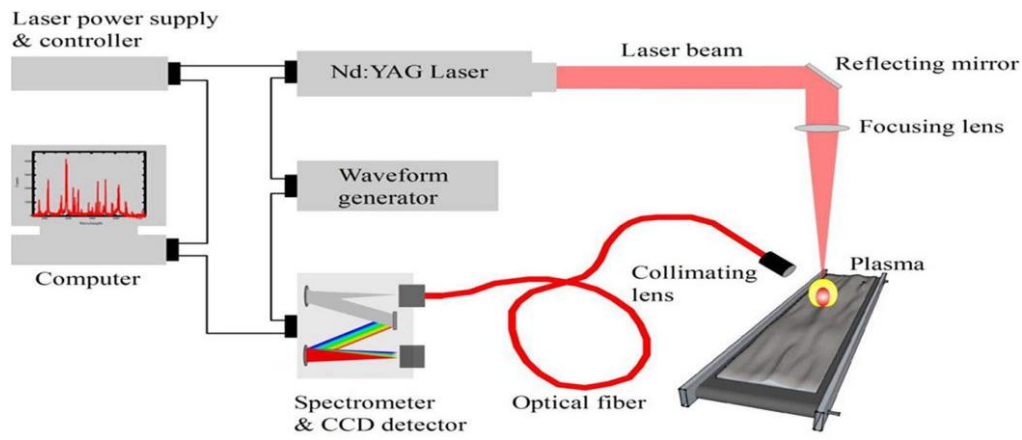


Figure 4.2.5: In a typical setup for laser-induced breakdown spectroscopy (LIBS) sample is placed above a platform. Nd-YAG: Neodymium-yttrium aluminum garnet [37]

4.3. Implementation of spectrophotometer

4.3.1. Hardware

The basis for putting together the initial prototype came from [37].

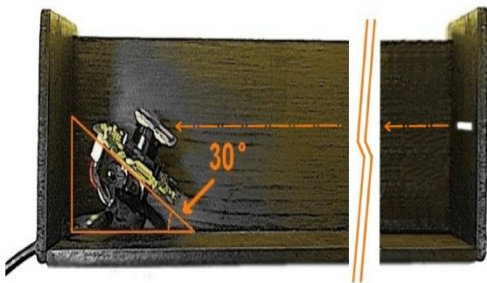


Figure 4.3.1.1: 1st prototype model

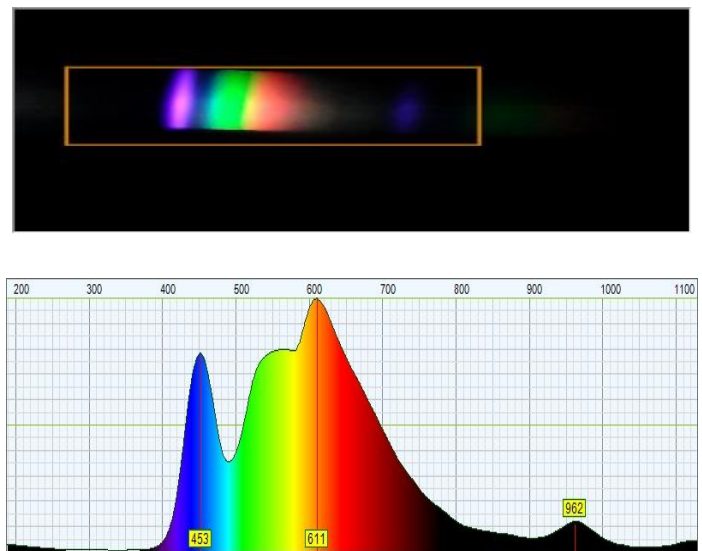


Figure 4.3.1.2: Results using Theremino Spectrometer software (fluorescent bulb)

The idea was to experiment with basic optical spectroscopy, of which there were many construction guidelines. Following the first implementation, it was soon discovered that for taking the spectral data of the constituent elements of a sample, an external method for exciting the sample was required. This method would depend on the physical form of the sample i.e. solid, liquid or gaseous. Based on the context of our project, in-situ analysis of geological samples, often found in nature as a solid form, requires a less intrusive approach which should not upset the environmental balance. The second prototype was designed based on a simplified LIBS model, incorporating the concept of using a powerful laser to vaporize a small part of the target sample surface.

The laser specifications were typical of a class 3B category model, which should be able to excite enough of the sample surface it hits to emit an observable spectrum by the camera. A convex lens of focal length 10 cm was chosen to focus most of the light, captured within the sample excitation area, towards the slit of 0.5 mm. The diffraction grating available for this project had 15,000 lines per inch. The diffracted light was directed 30 degrees away from the slit surface to ensure the camera readings were not interfered with by glare from direct light. Camera specifications.

Finally, the insides of the spectrometer were painted matte black to avoid any reflectance of light, as well as cutting down interference by stray light. The final setup model illustrated below

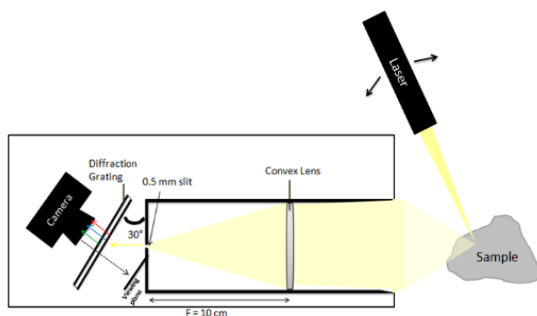


Figure 4.3.1.3: Spectrometer model design

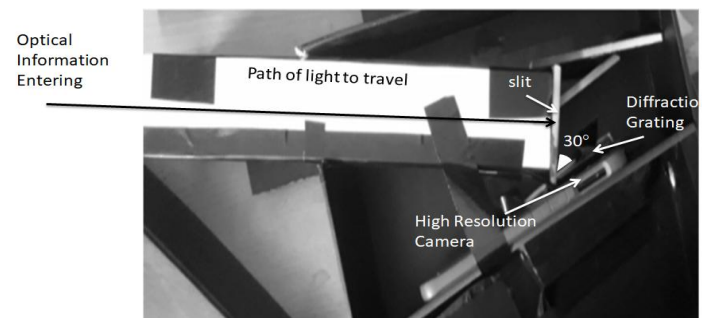


Figure 4.3.1.4: Final hardware constructed

The captured image, as well as real-time video feed from camera, can be fed through the Theremino Spectroscopy software which plots the spectral graph within the pre calibrated range. The spectral graph is further analyzed to extract the elemental information of the sample.

4.3.2. Software

The wavelength information from the spectral graph is further processed through an algorithm developed for this project, designed specifically to simplify the process of going through a pre-established database to match peak wavelength data to its elements. The program primarily focuses on the intensive search-based computation processes to aid in identifying the elements through its spectral signature.

To be more precise, the spectrum acquired from vaporizing a miniscule part from the surface of the sample under investigation will include a multitude of peak wavelengths. These wavelengths are a defining part of the emitted photons which contain information about the energy levels of constituent elements of the sample. Therefore, each element essentially contains a signature spectrum. The spectrum illustrates which of the characteristic wavelengths of the photons emitted are most frequent in number, in other words, which have the highest intensity during emission. Within a range of 200-1000 nm, the spectrum graph can offer over hundreds of such distinct photon information. So, for this purpose, this project prepared a focused dataset of spectral information for around 40 elements from the periodic table. These 40 elements are the most common to be found in nature, especially while studying geological samples. Moreover, there are detailed documentation available for each of the elements for the purpose of identifying them in extraterrestrial scientific analysis.

The source for the documentation used for this project is from the National Institute of Standards and Technology (NIST) LIBS Database. There is an additional open-source software, called the C-QuEST - ChemCam - Quick Element Search Tool, which provides a simplified database within its documentation. This software is directly referenced with NIST database, as well as elemental analysis from the Curiosity rover's ChemCam.

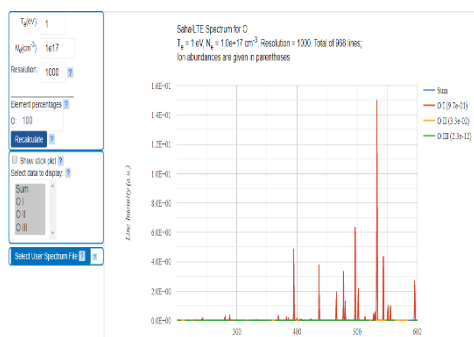


Figure 4.3.2.1: NIST database

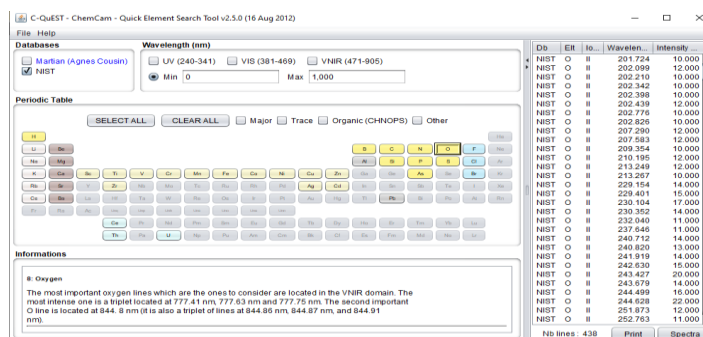
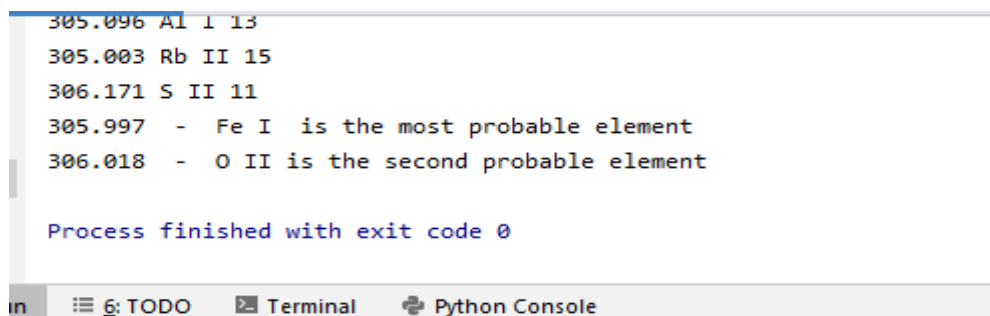


Figure 4.3.2.2: C-QuEST - ChemCam - Quick
Element Search Tool

The dataset, which has been preprocessed for the purpose of this project, contains the peak wavelength emissions of these 40 elements with the highest three intensity of occurrence. This is an essential part of narrowing down the results to the elements which are more likely to be found based on the peak wavelengths from the recorded spectra. The database is incorporated into the program as a .csv format to be accessed, which would allow incorporating more elemental data for future studies. The algorithm design consists of a basic binary search style structure where the peak wavelengths from the recorded spectrogram are fed individually as inputs to be searched. However, rather than focusing on finding a specific value within the database, a range of values are selected from the database which are considered to be the probable results.

Within the values of the selected range, the linear distance is calculated with respect to the peak wavelength, in order to determine the closest ‘neighbor’ within the two-dimensional stage. The output gives out the element which is most likely to correspond with the recorded wavelength, the second closest element it could be as well. The result is generated as follows



```
305.096 Al I 13
305.003 Rb II 15
306.171 S II 11
305.997 - Fe I is the most probable element
306.018 - O II is the second probable element

Process finished with exit code 0
```

Figure 4.3.2.3: Software calibrations and measurements

To attain further precision, the presence of the 2nd and 3rd most frequent intensities of wavelength are also checked within the rest of the recorded spectra. If one of both of this secondary condition is met, the higher the chances are of the element being present within the sample.

4.4. Results and Analysis

4.4.1. Data recorded

To test the accuracy of the final prototype, certain organic and inorganic based samples were placed under test. The test conditions involved conducting the experiment in a dark room to cut out any stray background light which would be treated as noise.

The spectrophotometer was first calibrated to record the spectrum with only the light sources (red, green, blue, white) to be used for testing and record the peaks and sharp dips along the x axis as background noise. The results are displayed as provided in the Figures below.

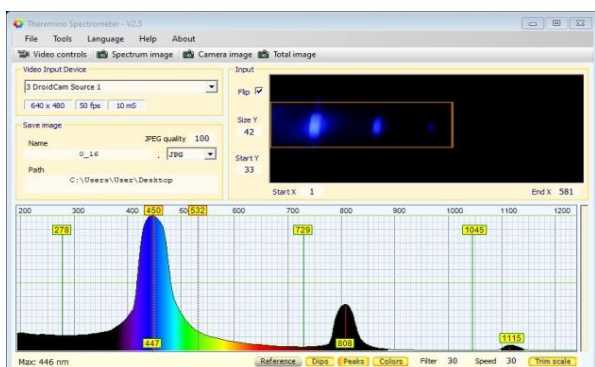


Figure 4.4.1.1: Blue light source

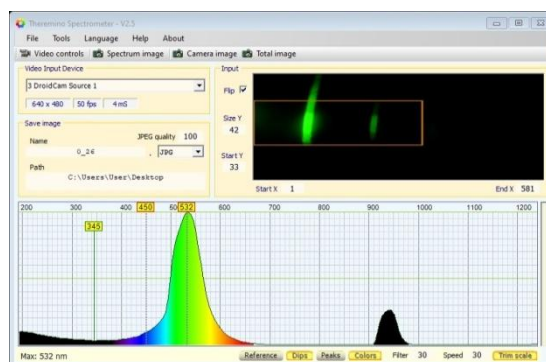


Figure 4.4.1.2: Green light source

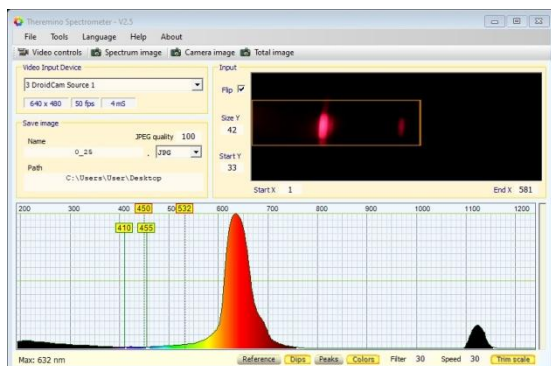


Figure 4.4.1.3: Red light source

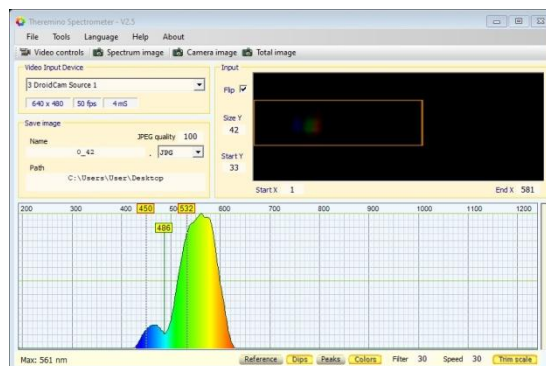


Figure 4.4.1.4: White light

Within standard room temperature, the samples were tested differently based on their physical form. Liquid samples were placed in a glass test tube and placed between the four different light sources at a time. The camera from the spectrometer picked up absorption spectra where the corresponding wavelength was recorded. The test was repeated 5 times with each light source and the commonly occurring wavelengths were run through the software to check for probable matches within the database.

A different procedure was followed for the solid sample testing. Points on the sample were exposed to a concentrated heat source while each of the light sources illuminated the sample surface placed under investigation. The values were recorded the same way as the liquid samples. The following table illustrates the comparisons between the test results and the expected results from different certified spectroscopy test sources:

Table 4.4.1.5: Recorded results to test the spectrophotometer design

Sample	Expected Results	Actual results	Characteristic Wavelength detected (nm)
Coconut Oil	C, H, O	C, O	723, 345
Olive Oil	C, H, O	O, C	345, 493
Water	H, O, 7 other common dissolved metals (eg. Fe, Zn, Cu etc)	O, Fe, Cl, Mg	345, 373, 858, 999
Leaf	C, N (9 other elements)	C, N, Fe, Mn, O	274, 280, 249, 280, 345
Flower Petal	C, N, O	C, O	274, 726

4.4.2. Analysis and discussion

The effectiveness and accuracy of the spectrophotometer design should ideally be measured against its performance in a simulated environment of extraterrestrial terrain it was intended for testing on. There are many other variables to be considered that could contribute to the data collected to be corrupted or be overwhelmed by noise (background light). For instance, the atmospheric condition for outdoor testing would be very different from the current set of results taken indoors.

Moreover, an extraterrestrial planet would inevitably provide other environmental factors which scientific equipment tested on Earth would be unprepared for. However, despite the limitations of diverse testing conditions, the initial tests to verify the operational effectiveness and accuracy of the designed spectrophotometer can be summarized as follows:

Table 4.4.2.1: Interpreting the results

Sample	No. of Correct/No. of test elements	Accuracy (%)	Remarks
Coconut Oil	2/3	66.67	Relatively accurate
Olive Oil	2/3	66.67	Had problems identifying carbon properly
Water	4/9	44.44	For the sample of local water, the most dominant elements were precisely identified
Leaf	5/11	45.45	Unclassified leaf sample, main components identified
Flower Petal	2/3	66.67	Unclassified flower petal sample, reference source not very detailed

The key observations are noted below:

- From the selected reference sources, the major constituent elements were prioritized over any other metals that may be present in the sample based on its origin, class or category. Specifically, the identification of Carbon and Hydrogen could indicate the presence of molecules within the hydrocarbon family but it cannot be deduced for certain through the initial method of testing done for the project. Samples were tested under the assumption that they were “pure”.

In reality, these samples contained foreign elements and compounds that did in fact interfere with the recorded results. However, the foreign elements were not included as part of the result since this study does not include measuring the purity of samples, which is better conducted under sophisticated laboratory conditions.

- The identification of Oxygen in most of the samples could possibly mean two things -
 - i) There were Oxide-rich compounds in the samples
 - ii) The test environment had a higher Oxygen concentration which interferes with the results.
- The leaf and flower petal samples were initially considered to be the best indicators samples for organic compound presence. While Carbon and Nitrogen were significant findings, the analysis could not deduce if organic or protein molecules are present in the sample. No readings were able to identify Hydrogen as a constituent element of the samples.

4.5. Design review/ summary

The initial tests conclude that the designed spectrophotometer was able to identify some of the constituent elements of liquid and soft solid samples, with the latter needing external excitation source, that are present in high concentration. It cannot detect directly any trace of organic compounds within the samples which is a crucial part of investigating extinct or extant life on any planet. With further tests in more dynamic test conditions this design could be improved. The software database needs to be updated with information on different families of organic and inorganic compounds which would enable matching “fingerprint regions” of the resultant spectrum.

Chapter 5

Complete Prototype Design and Deployment

5.1. Introduction

In reference with all the aspects and terminology covered rendering throughout the project a prototype rover bot was deployed to illustrate all the processing and a guide to the pipeline of all the implementations. In order to demonstrate the prototype, using a poly board as a base for the arrangement with the Spectrometer and Raspberry Pi with the Coral USB Accelerator in line with motor gears and Arduino for movement was constructed. The finalized layout is given with schematic designs in the following sections of this chapter.

5.2. Design and Development of the Custom Prototype Body

Before putting all the components together and merging to form the desired model prototype the design schematics were configured with the mathematical calculations regarding movement, turnarounds and calibrations. The approached models were designed with calculating factors for mechanical movements and hovering of the bot within the aspects of the area of interest. Also, power distribution of the components was taken before finalizing the model setup so that, risk factors concerns would be avoided.

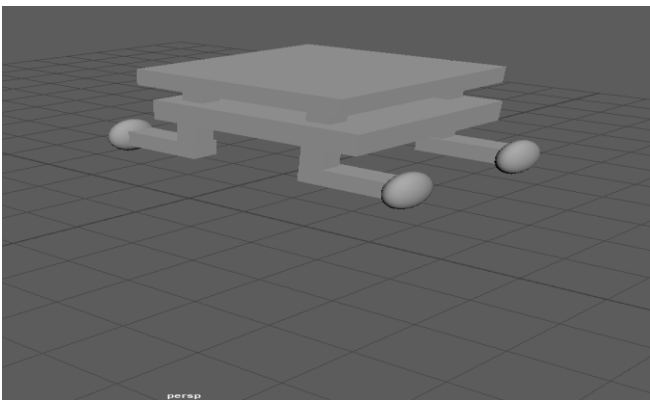


Figure 5.2.1: Approached Models Schematic Designs (Perspective View)

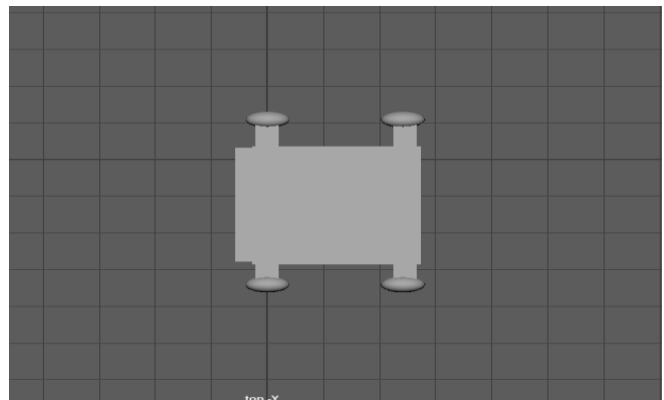


Figure 5.2.2: Different views of the designed prototype (Top View)

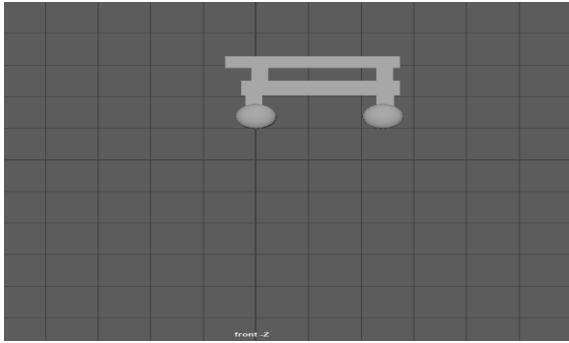


Figure 5.2.3: Different views of the designed prototype (Front View)

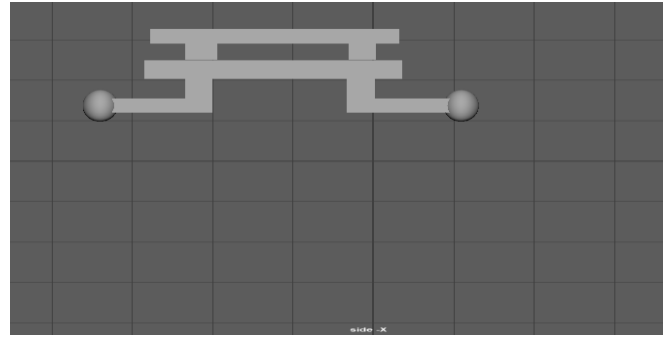


Figure 5.2.4: Different views of the designed prototype (Side View)

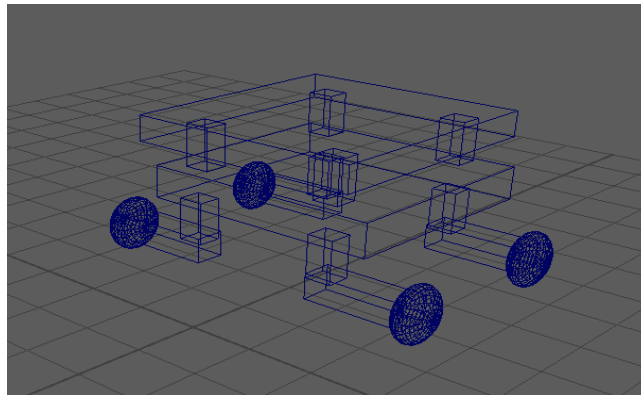


Figure 5.2.5: Approached Models Schematic Designs (Wireframing)

5.3. Finalized Design and Deployment

After setting the hardware with all the necessary components the finalized prototype design, which was presented for demonstration, is illustrated in the Figures below.



Figure 5.3.1: The finalized hardware setup prototype body (view 1)

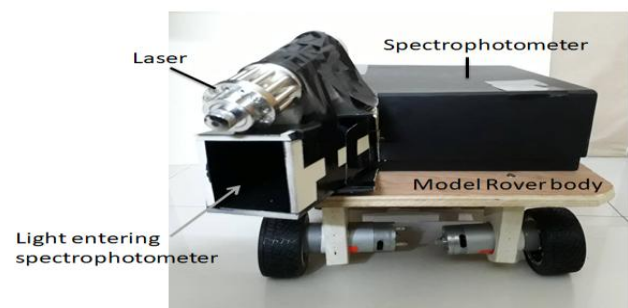


Figure 5.3.2: The finalized hardware setup prototype body (view 2)

5.4. Robot at Work

For operating of the robot, it has a few sequences attached to it. First of all, it will start moving when power is provided and roam around in 360 degrees to view the surrounding environment. From the generated video buffer or frame captures it will identify its point of interest and move to its location through mechanical movement. When it finds the point of interest it will do the image processing consisting of segmentation and classification based on the given custom classified model data set built-in and identify the objects based on its ability to find the specified rock and map out the materials or formations associated with it. Then, the spectrometer will create and provide the illustration given in the figure below to demonstrate the scientific analysis based on the detected object. Finally, based on the frames fed into the buffer it will prop out the detected objects and provide classification of which rock is which and do a scientific analysis.

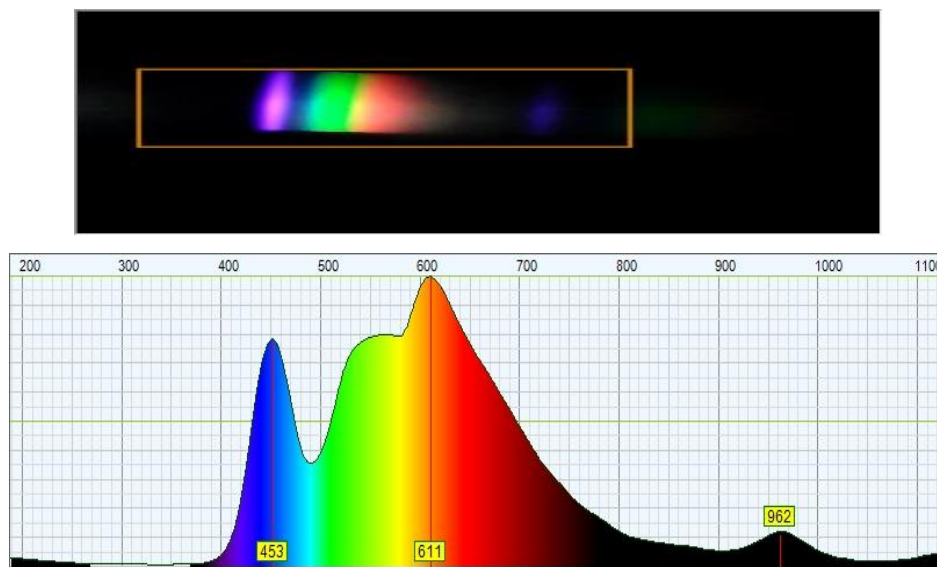


Figure 5.4.1: An illustration of the Scientific Analysis performed in the spectrometer

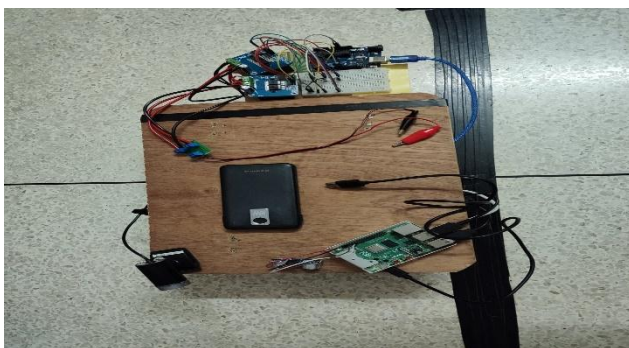


Figure 5.4.2: The Finalized Prototype Setup (1)



Figure 5.4.3: The Finalized Prototype Setup (2)

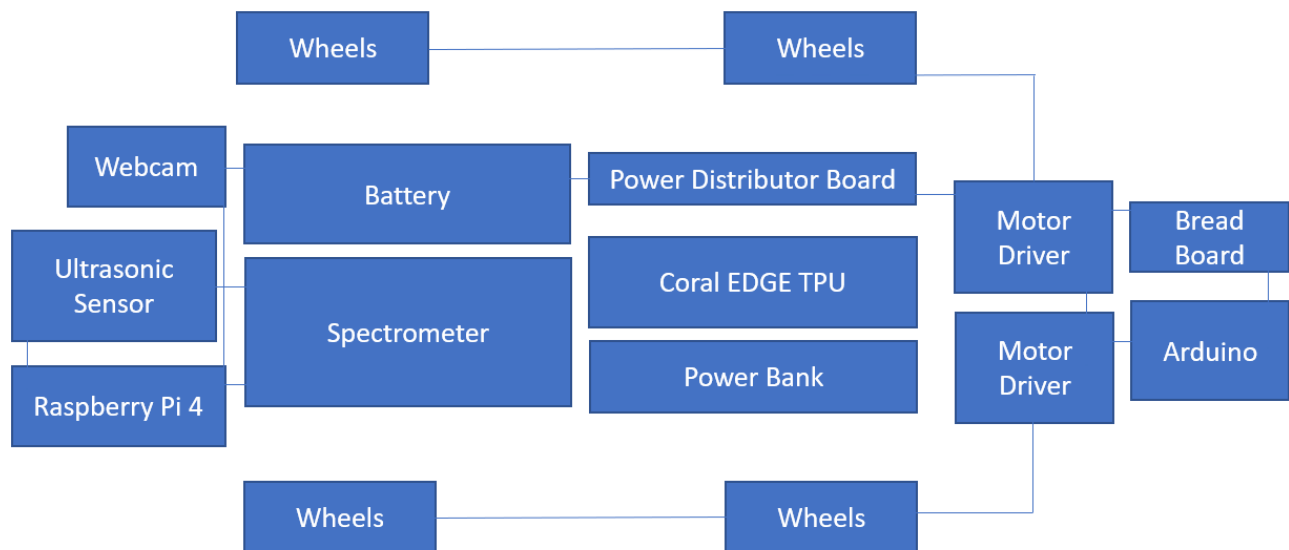


Figure 5.4.4: Flowchart of the designed prototype



Figure 5.4.5: Finalist version designed prototype

Chapter 6

Experimental analysis with quantum computing

6.1. Introduction

While classical computing devices have evolved following the Turing model, there has been much debate about efficiency with regards to what computers were originally set out to do. Alan Turing's machine was based on the concept of being able to solve any math, making mathematicians job easier; machines that can run algorithms given the right mathematical program. But this model cannot take on calculations derived from quantum mechanical models. Models which are crucial to understanding nature and most of the real physical world. Later on, David Hilbert would go to define that a more 'natural' computing device should also be able to simulate quantum-mechanical processes.

6.2. Methodology

The concept for this part of the study initially considered three different types of algorithmic and programming concepts.

1. Comparing basic search algorithms

One of the most anticipated features of quantum computing is that it can speed up search-based algorithms. Given an array of n numbers, the worst-case scenario for searching one number within that array are n computational iterations. If n proved to be close to a billion numbers, the classical computer will possibly need close to a billion iterations as well. Mathematically, quantum computers provide a much better solution - even in the worst case, the algorithm will take $n/2$ iterations which is significant improvement in terms of computational overhead.

Some of the algorithms that have been theoretically designed for testing search operations in quantum computers, such as Grover's algorithm or Deutsch-Jozsa algorithm, mathematically prove the speed-up effect quantum computers can have. Since there are no error corrected quantum computer available yet, these serve as a basis to prove the accuracy of any model that may be developed in the future.

This process, however, cannot be substantially proven through any traditional input and output methods, therefore the scope of this project could not cover this aspect.

2. Simulating the chemistry of elements and molecules

An exciting way in which quantum computers can revolutionize our understanding of the physical world is if we can simulate the natural processes through the pattern of chemical interaction between molecules. All matter in existence came from the constituent elements, which evolved from combining similar atoms, and subatomic particles preceding their evolution. The idea for this kind of study was to understand the depth of behavior to the very lowest energy state of an atom.

While at such small scale, the traditional methods of measurement do not apply to near-quantum scales, quantum computers can display, through the probabilistic randomness of its own nature, how atoms can interact based on energy level excitation. This aspect of the computing process, again, could not be demonstrated using the limited computational resources available for this project.

3. Generating ‘True’ Randomness

This project, therefore, explores the concept of generating random numbers through quantum-mechanical processes. In reality, classical computers cannot generate “true” random numbers. Even the most dynamically efficient high-level language libraries rely on ‘seed’ specification from the user to generate seemingly random numbers. This process works well for the limited scope of applications we usually deal with. But the underlying concept of these traditional processes utilizes a formula that iterates on itself.

A qubit, on the other, follows the quantum-mechanical rule, where in a state of superposition, there is a probability it could either be a 0 or a 1 at a certain time of measurement. The idea can be represented as follows mathematically: $\alpha|0\rangle + \beta|1\rangle$ Where α and β could be any probability between 0 and 1.

This randomness could be utilized to simulate scenarios for applications that require high level of testing. The reason this could be very useful for space exploration is because of the large number of variables that the spacecraft and rovers should be tested for. Coupled with the computational speedup, with very low number of qubits actually needed to perform these calculations, the range for mathematical testing could be increased exponentially.

6.3. Implementation

6.3.1. Circuit Model

The initial circuit model tested on two qubits:

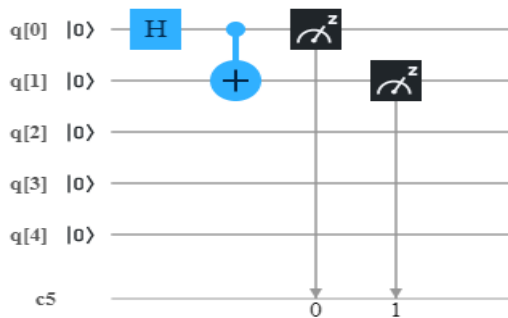


Figure 6.3.1.1: First test quantum circuit model

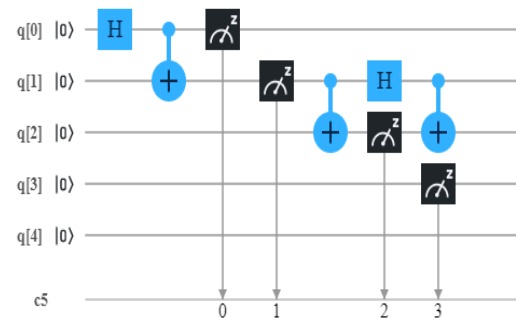


Figure 6.3.1.2: Second test quantum circuit

On an available setup of 5 qubits, the initial qubit in the first row is setup as $|0\rangle$. A Hadamard (H) gate is applied to it which creates a state of superposition mathematically represented as $H|0\rangle = (|0\rangle + |1\rangle)/2$. Next, a CNOT (Controlled-NOT) gate is applied which works similar to an IF condition in classical programming. There is a control bit which determines what the state of a target bit will be. For example, if the control bit is $|0\rangle$, there will be no change to the target bit. But if the control bit is $|1\rangle$, the target bit reversed like a classical NOT operation. There are composer references to measure the value of each bit after the gates are applied. The second test circuit increased the number of gates with the same pattern to generate more ‘random’ qubit state using the H gates.

6.4. Results and Analysis

6.4.1. Data recorded: IMB Quantum Computer Results

Two sets of results were taken by running the same circuit program twice. The results are illustrated as follows:

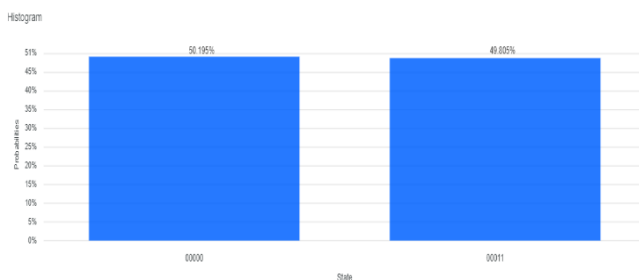


Figure 6.4.1.1: First Circuit (using two qubits) (1)

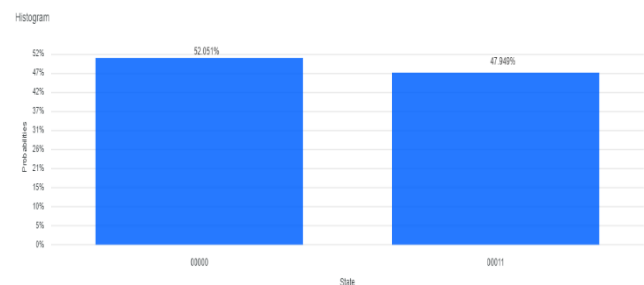


Figure 6.4.1.2: First Circuit (using two qubits) (2)

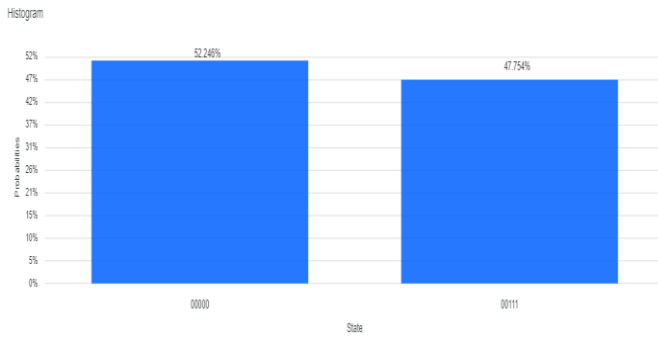


Figure 6.4.1.3: Second Circuit (three qubits) (1)

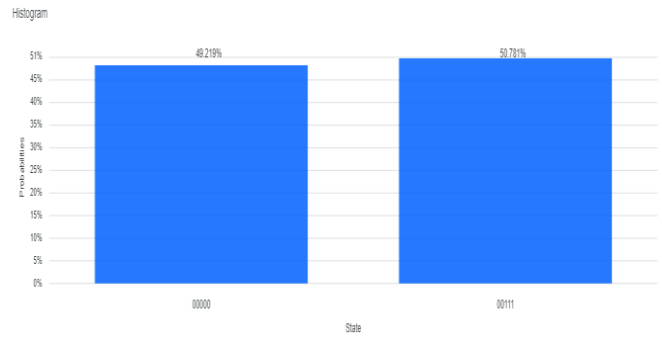


Figure 6.4.1.4: Second Circuit (three qubits) (2)

6.4.2. Analysis and discussion

The results show that the state of $|0\rangle$ and $|1\rangle$ are completely random at output. There is no sure possibility that at a certain time the qubit can be measured as an absolute 0 or 1 (classical computation measurement). The H gate converted the qubit state as any possibility between $|0\rangle$ and $|1\rangle$, therefore when the CNOT gate chose a state for the control bit, the state with the most probability at that exact time of measurement was chosen. That is why, the state of the target bits at the end measurement were interpreted within a percentage range of how likely the output will be measured as 00 or 11 for two qubits or 000 or 111 for three qubits. Since the same circuit was run twice through the IBM free access quantum computer, under the exact same conditions, the varying percentage of results further justify the randomness of the state measurement and what consecutive result we get at the end to work further with.

6.5. Design review/ summary

Currently, without having a strong idea about the mathematical nature of qubits, it is relatively tough for the usual consumer to understand the effectiveness of a quantum computer. In fact, unless there is a specific mathematical factor which would require exponentially large computational resources, the average consumer does not even need to break out of the classical computation model to satisfy regular programming needs.

Chapter 7

Conclusion

7.1. Summary of Findings

From the three different sets of result analysis in accordance to the three major aspects of this project, the concluding remarks could be summarized accordingly:

- 1) The deployed image processing is capable of performing both image detection and segmentation for a particular image set given the scope of rocks. The 42 categorical segments illustrate that the deployment mechanism is capable of detecting rock objects provided any rocky surface area (illustration based on the images taken from MARS). Also, one more distinction to point out up till the scope of research, no datasets concerning rocks was present and not available under any development site or scopes of projects based upon the approached terminology. Also, the provided sequences online had compatibility and usage activity issues which was resolved and updated with the development given our aspects regarding our project.

The processing was also calibrated and updated for the real time render in video feeds with buffers however, the outputs were not optimum and proficient in terms of the desired results based on the tasks provided to the system. It was also experimented with the trademark benched peripheral devices such as the Google Development Board and the renowned Nvidia Jetson Nano Developer Kit. Although, for the finalization concerning the book was bounded to the Coral USB Accelerator with the Raspberry Pi 4 Model B.

- 2) The designed spectrophotometer is capable of identifying some of the major constituent elements from the sample under test conclusively. However, trace elements cannot be identified with confidence, unless the method of exciting the sample externally is amplified. This is important because extracting information from even the slightest trace of elements within a geological sample can contribute to important discoveries in finding extinct or extant life in a foreign planet. The scientific equipment is required to be sensitive to a high level in order to give conclusive results about the scientific objectives.

It should be worth noting that only for testing purposes, a diverse array of samples in two different physical forms were placed under investigation initially. Spectrophotometry for liquid samples work differently than for solid samples. To test the accuracy of the design, both types of samples were investigated, whereas the core concept of the project ideally deals primarily with solid samples available undisrupted in nature.

The test however used appropriate methods of exciting and extracting the optical information from the samples based on their physical form. Therefore, to truly test the effectiveness of the designed spectrophotometer for the context defined, which is derived from a typical LIBS set-up for investigating solid samples, there should be further improvement into the external excitation method of the geological samples such as rocks. The second set of results aims to show improvement based on this standard through using a more powerful laser to excite the samples.

- 3) The initial experiments conducted through the available quantum computer at IBM aims to test the basic understanding of quantum-mechanical phenomenon through the quantum gates currently available. Quantum computing is still at its infancy, with error-corrected models yet to be available.

The initial tests prove the theory of potentially attaining ‘true’ randomness through the superposition phenomenon. Further down the line this attribute can be useful for conducting diversified tests for space exploration purposes.

The mathematics for quantum search algorithms also proves the exponential increase of computational power through very less amounts of gates compared to the scale of gates required for classical computers.

- 4) Overall, from the scope of this project, it can be proved that for the field of space exploration, quantum computing can lend a massive advantage. However, quantum computers will not lend any advantage for the average consumer, where there is no need for high scale computational power.

7.2. Overall Project Finance

Table 7.2.1: Image processing and spectrometer costings

Components	Price (BDT)
Raspberry Pi 4 model B (4GB)	6500/-
Coral USB Accelerator	13,477/-
Webcam/ High Resolution Camera	2500/-
Power Adapter, Display Port etc.	500/-
Micro SD Card (32GB)	500/-
Diffraction grating	1500/-
Convex lens (f = 10 cm)	250 x 2/-
Board, black tape, matte black paint etc.	500/-
Total	25,977/-

7.3. Novelty of the work

The project not only focused on new and experimental concepts in the domain of scientific exploration of extraterrestrial terrain, but also grasped the basic understanding of quantum computing. The precedent set by the work from [1,2,3] and [2,3,4] have inspired the geological image processing aspects of this project, which have yet to be replicated for terrestrial terrain.

The research domain for image detection and segmentation is more useful in the context of general consumers. Naturally occurring geological scenes are rarely of interest which can be attributed to the very low number of research papers available for prioritizing rock segmentation, detection and identification. Furthermore, the spectrophotometer design is unique from the perspective that, while this has been derived from a generic Do-It-Yourself (DIY) model for analyzing light spectrum, the setup with the addition of the lenses enables more sensitive optical information to be captured by the camera.

From the overall results of the spectrophotometer, it can be deduced that this modification is applicable for samples of different physical states, more dynamic among the cheaper models and a new form of LIBS setup. The quantum computing aspect of generating ‘true’ randomness space exploration testing is a completely new hypothesis that has been set for this project. Most researchers have been focusing on experimenting with different ways quantum circuits can be of advantage to certain applications, however, all these findings will end being unique.

The current need for randomness arises from the demand of efficient and powerful encryption generation. However, this theory is further explored in the project in terms of testing for space exploration.

7.4. Final Impact of the project

The studies conducted for this project shed light on many aspects of the technology that are currently deployed on space exploration rovers. The impact this technology could have for us upon further development in the future are summarized in the following sections.

7.4.1. Survey on Environmental Impact

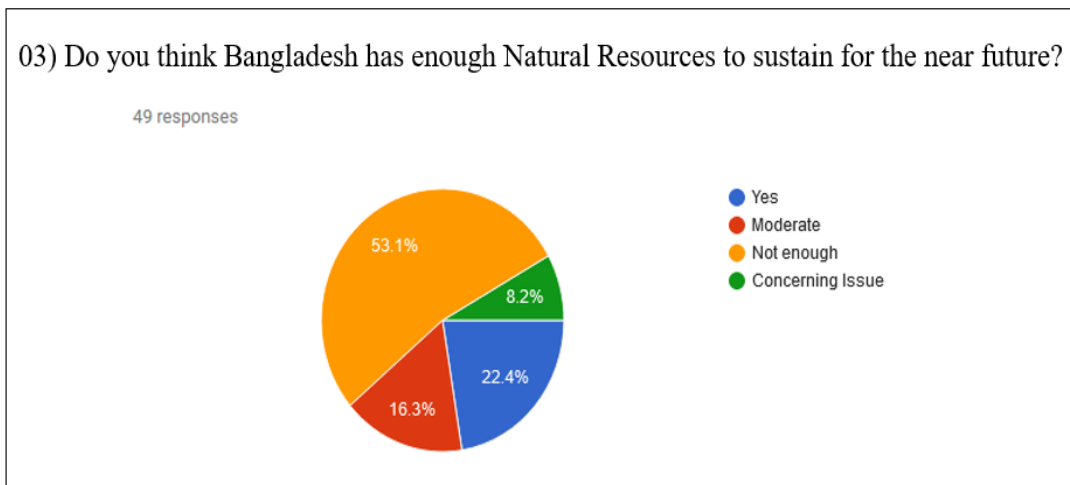


Figure 7.4.1.1: Overview of responses for question 3

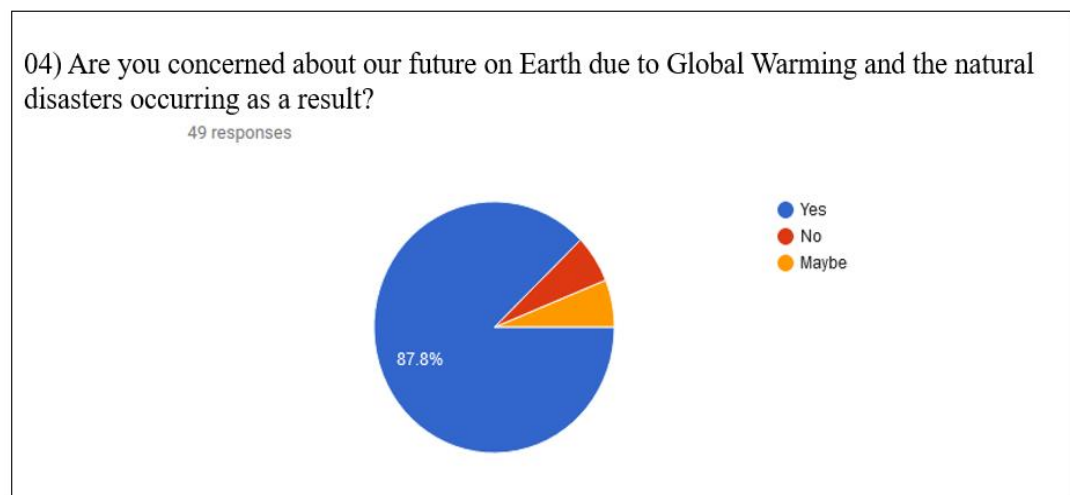


Figure 7.4.1.2: Overview of responses for question 4

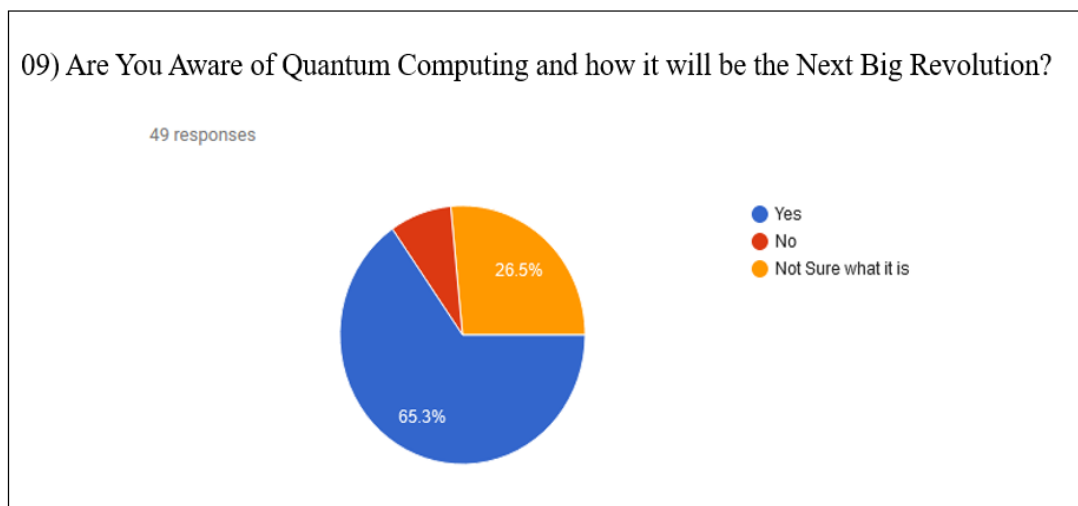


Figure 7.4.1.3: Overview of responses for question 9

This survey picks up from the same group of participants from section 1.5. The studies from this project are relevant for people who understand the need for exploring rover-based technology.

7.4.2. Future scopes

The scope of this work is far reaching and can be used to diversify the working capabilities of rovers which would enable remotely conducting research in environments deemed too dangerous for humans to explore physically. However, this project has been limited to explore only specific domains of computer vision and image processing to detect rocks, navigation capability only as intelligent as interpreting basic GPS navigation and relative positioning. The hardware will be designed to accommodate the spectrophotometer, which will be calibrated mathematically to give an estimation of geological value (focusing only on basic elements), and the Raspberry Pi that houses the processor intended to be used for this project.

7.5. Limitations

One of the main challenges for this system would be the verification of unknown geological phenomenon that may occur in the case of extraterrestrial terrains. It would be fairly realistic to test the system for Earth based geology but the performance for its intended purpose would be limited to test. This project also estimates the use of various computational domains such as basic machine learning concepts, pattern recognition between elemental data and mapping in-depth data from small geologic areas to put together a bigger, cohesive visual model. The current existing algorithms facilitating this framework are not fully error corrected yet.

The research on these domains is still progressing towards further improvement; therefore, it's hard to deduce, to a certain degree, the accuracy of the system being developed. It also takes a long time to train the system to identify patterns and relations. Geological sensors appropriate for any in-depth elemental or mineralogical studies have proven to be expensive and industrial grade complex machines. Developing a specialized machine that would fully meet the standards for such studies in geological fields is out of bounds for this current project. Therefore, any 'scientific' sensors used would be fairly limited and appropriate for rough estimations about the geological value of a certain area.

7.6. Ethical concerns

One of the biggest challenges this project faced was the testing of the scientific equipment. The ethical issue is not to contaminate the environment from where the samples for investigation are to be taken. Effective in-situ analysis often calls for extremely intrusive sample extraction processes which might upset delicate ecological systems. Throughout the research there have been many instances where the powerful lasers have been used. Lasers of class 3B used in this project are extremely dangerous to humans upon the slightest exposure. Testing had to be conducted very carefully through shielding the area where the sample is being burned by the laser.

7.7. Conclusion

Humanity will always strive to explore beyond Earth, embark on expeditions which only seem impossible today. This project not only allowed the team to explore what such a future might hold through technological leaps in the near future but also provided a window into the very depths of a scientific world most of us often cannot comprehend. It explores the hopes of humanity to find life beyond this planet and the incredible feats of engineering this ambition has led us to today.

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Appendix A

Survey Questionnaire

Survey Questions (Project_Thesis_Group_28)

Please give a Tick Mark on your choice of answer

01) Which Age group do you belong to?

[A] 15-20 [B] 20-25 [C] 25-30 [D] 30 and above

02) Which Profession do you belong to?

[A] Student [B] Teacher [C] Job/Service holder

03) Do you think Bangladesh has enough Natural Resources to sustain for the near future?

[A] Yes [B] Moderate [C] Not Enough [D] Concerning Issue

04) Are you concerned about our future on Earth due to Global Warming and the natural disasters occurring as a result?

[A] Yes [B] Maybe [C] No

05) Have you heard about space agencies like NASA or aware of the work they do?

[A] Yes [B] No

06) Should we explore other planets like MARS to determine if humans can live on them?

[A] Yes [B] No [C] Space Travel sounds cool

07) Do you think space exploration to find alien life and human habitability potential is only possible in sci-fi movies like 'Avengers' or 'Interstellar'?

[A] Yes, it will only waste money in reality [B] No, I think we'll achieve this in reality

08) Is it necessary for Bangladesh to invest in scientific exploration of extraterrestrial planets?

[A] Strongly Agree [B] Agree [C] Not Sure [D] No Need

09) Are You Aware of Quantum Computing and how it will be the Next Big Revolution?

[A] Yes [B] No [C] Not Sure what it is

If you have any comments or queries please state below:

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